



openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

Signing Binaries on OBS

Introducing pesign-obs-integration

Joey Lee <jlee@suse.com>
SUSE Labs engineer

Yogyakarta | October, 2026

Agenda

- The architecture for signing a binary on OBS
- How to use pesign-obs-integration
- Some details in pesign-obs-integration
- Two signing format examples

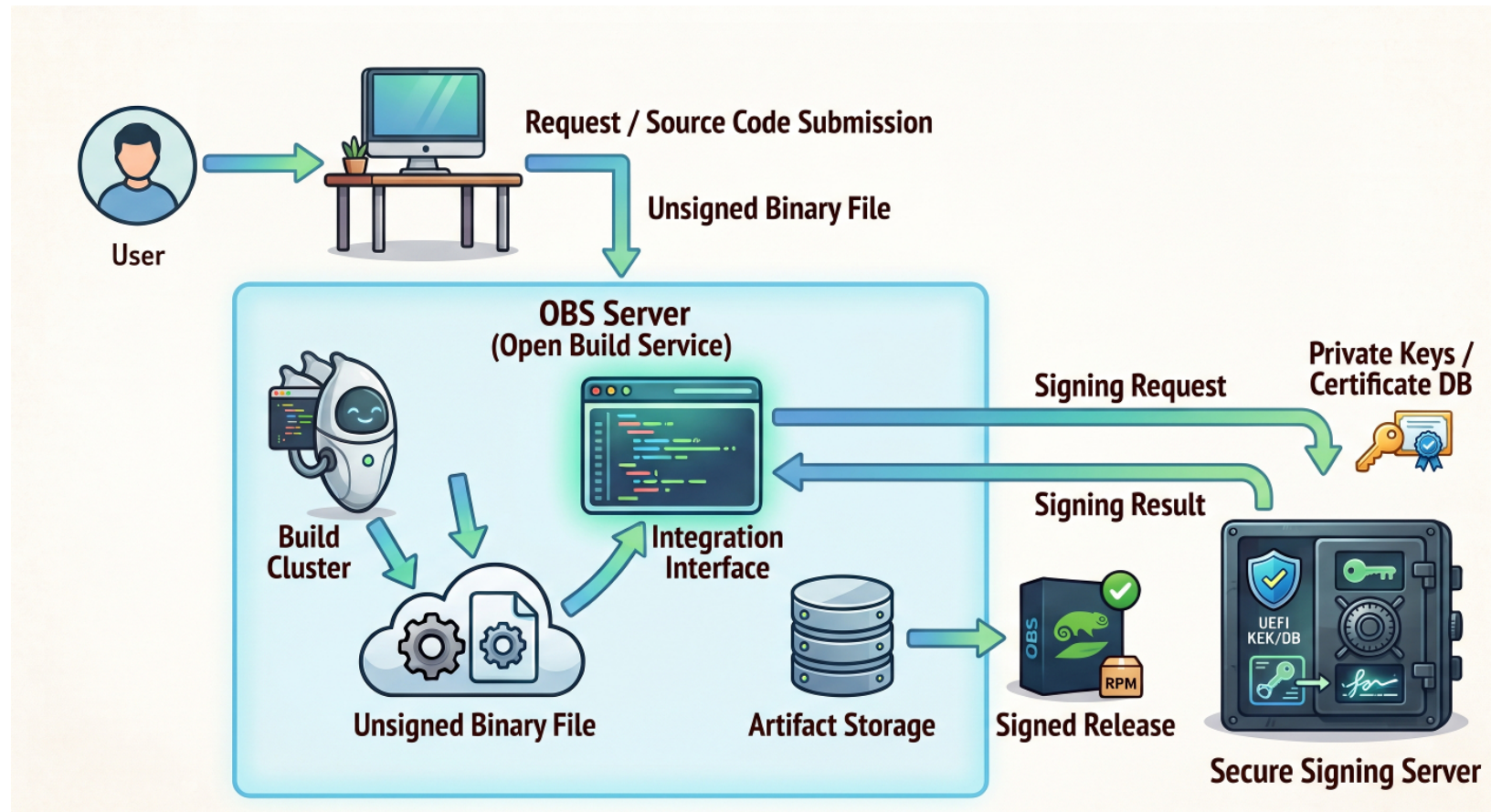


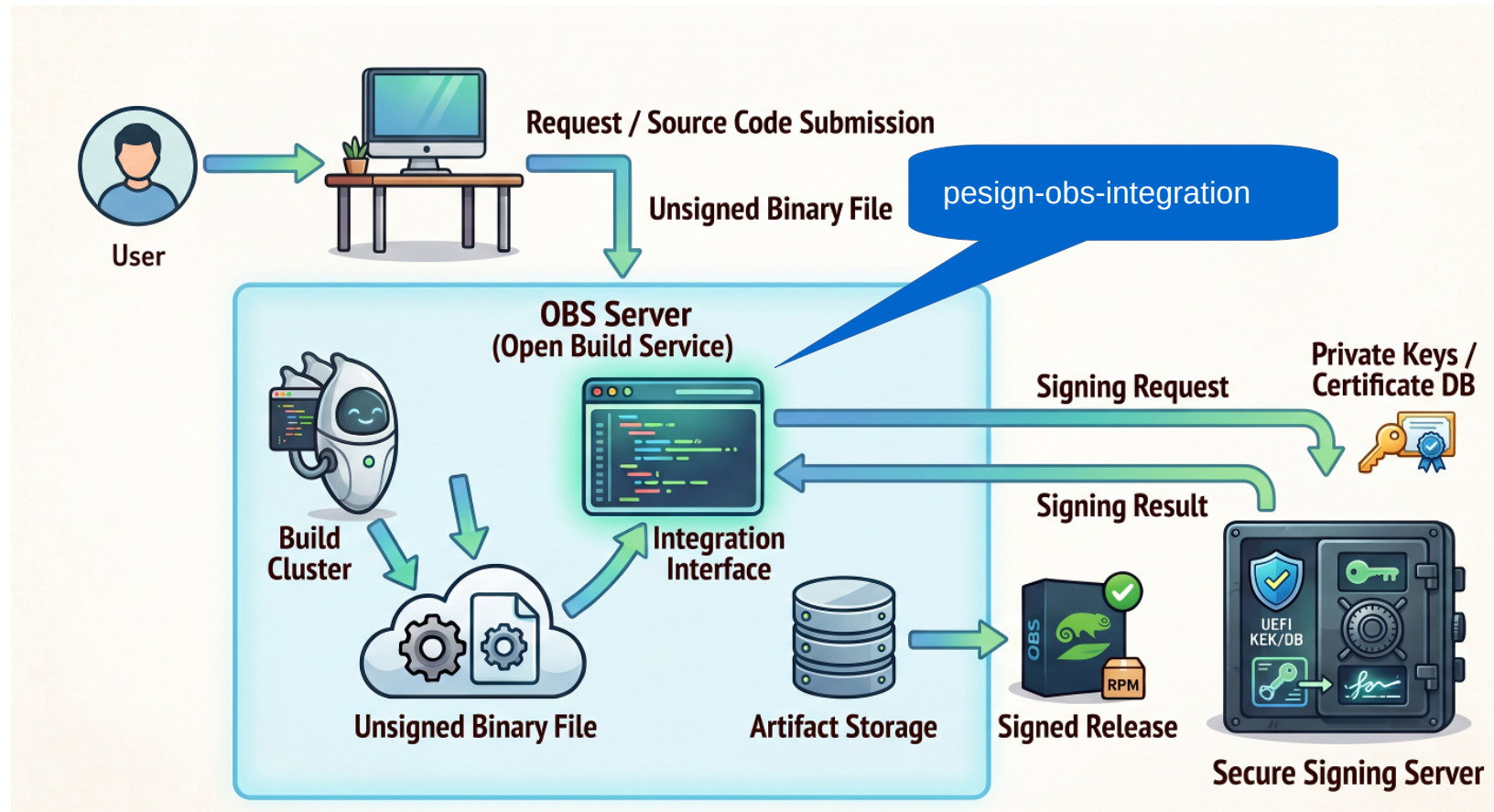
openSUSE®

Asia Summit 2026

Yogyakarta, October 3 - 4

The architecture for signing a binary on OBS







openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

How to use pesign-obs- integration

Using pesign-obs-integration

In your .spec file:

- # needssslcertforbuild
- export BRP_PESIGN_FILES='pattern...'
- BuildRequires: pesign-obs-integration

Using pesign-obs-integration

In your .spec file:

- # needssslcertforbuild
- export BRP_PESIGN_FILES='pattern...'
- BuildRequires: pesign-obs-integration

KMP (Kernel Module Package) *.ko builds trigger pesign-obs-integration automatically.

Supported binary format

- *.ko
- efi binaries
- ESL (EFI_SIGNATURE_LIST)
- *stage3.bin
- grub.elf

Corresponding file name

- *.ko

 - *.ko|*.mod**

- efi binaries

 - /boot/* | *.efi | */lib/modules/*/vmlinu[xz] |**

 - */lib/modules/*/[li]mage | */lib/modules/*/z[li]mage**

- ESL (EFI_SIGNATURE_LIST)

 - *.auth**

Corresponding file name

- *.ko

Examples: vhba-kmp, virtualbox:kmp

.ko|.mod

Examples: shim, grub2

- efi binaries

/boot/* | *.efi | */lib/modules/*/vmlinu[xz] |

/lib/modules//[[li]mage | */lib/modules/*/z[[li]mage

- ESL (EFI_SIGNATURE_LIST)

Examples: kernel-default

*.auth

Corresponding file name (cont.)

- ***stage3.bin**
 - 028e04a32e61 sign stage3.bin from s390-tools with
- ***grub.elf**
 - 13efe2232909 Support ppc grub signing

BRP_PESIGN_FILE ex.

- *.ko

- *.ko|*.mod

- efi binaries

```
/boot/* | *.efi | */lib/modules/*/vmlinu[xz] |
```

```
*/lib/modules/*/[li]mage | */lib/modules/*/z[li]mage
```

- ESL (EFI_SIGNATURE_LIST)

- *.auth

```
BRP_PESIGN_FILES="%modules_dir/%image"  
BRP_PESIGN_FILES="$BRP_PESIGN_FILES *.ko"  
export BRP_PESIGN_FILES
```

BRP_PESIGN_FILE ex. (cont.)

- *.ko `export BRP_PESIGN_FILES='KEKUpdate_openSUSE_PK.auth'`

`*.ko|*.mod`

- efi binaries

`/boot/* | *.efi | */lib/modules/*/vmlinu[xz] |`

`*/lib/modules/*/[li]mage | */lib/modules/*_image`

- ESL (EFI_SIGNATURE_LIST)

`*.auth`

Advanced information

- <https://github.com/openSUSE/pesign-obs-integration.git>
- The "# needssslcertforbuild" comment tells the buildservice to store the signing certificate in **%_sourcedir/_projectcert.crt**
- The brp-99-pesign script is a start



openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

Some details in pesign-obs- integration

Execution flow

Main Logic on OBS

brp-99-pesign
(bash)

Logic Template

pesign-repackage.spec.in

pesign-gen-repackage-spec
(perl)

Execution flow

Main Logic on OBS

1. First
rpmbuild



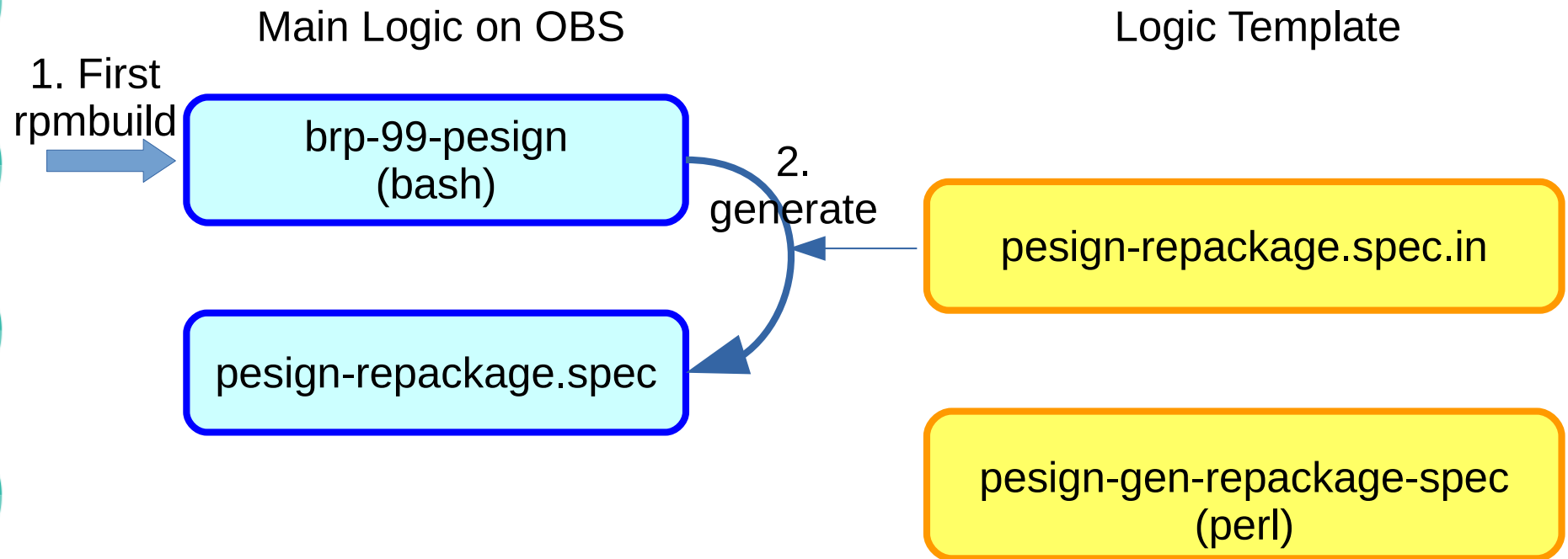
brp-99-pesign
(bash)

Logic Template

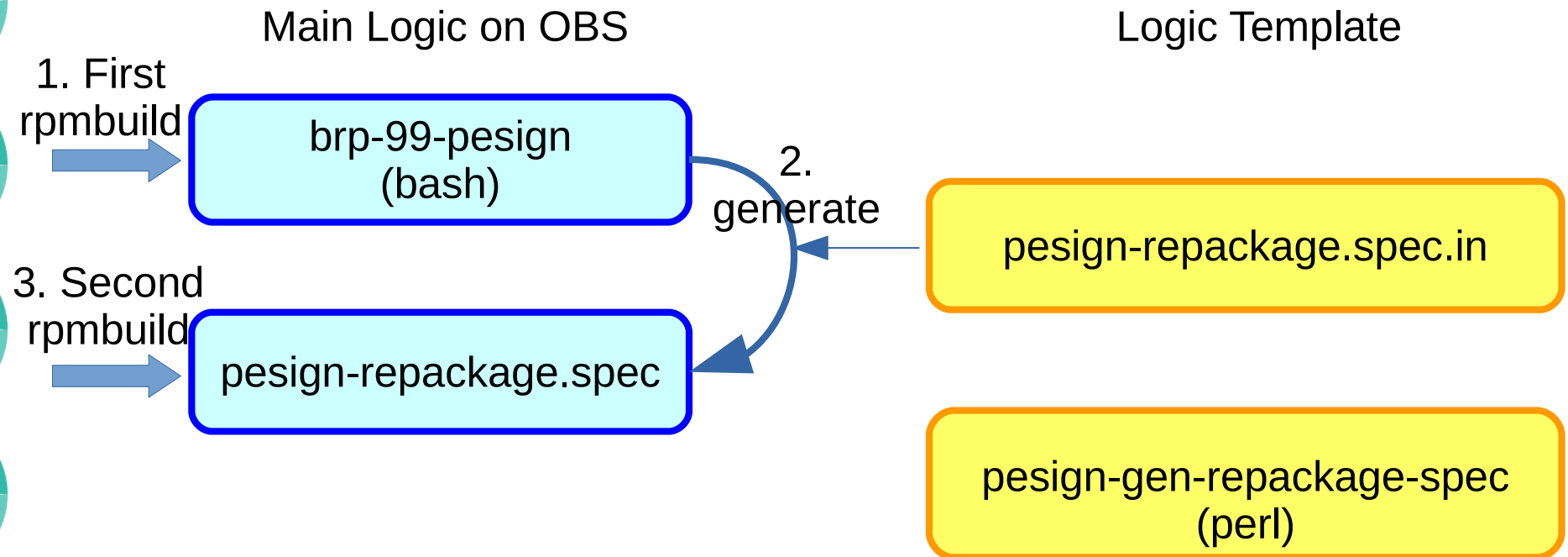
pesign-repackage.spec.in

pesign-gen-repackage-spec
(perl)

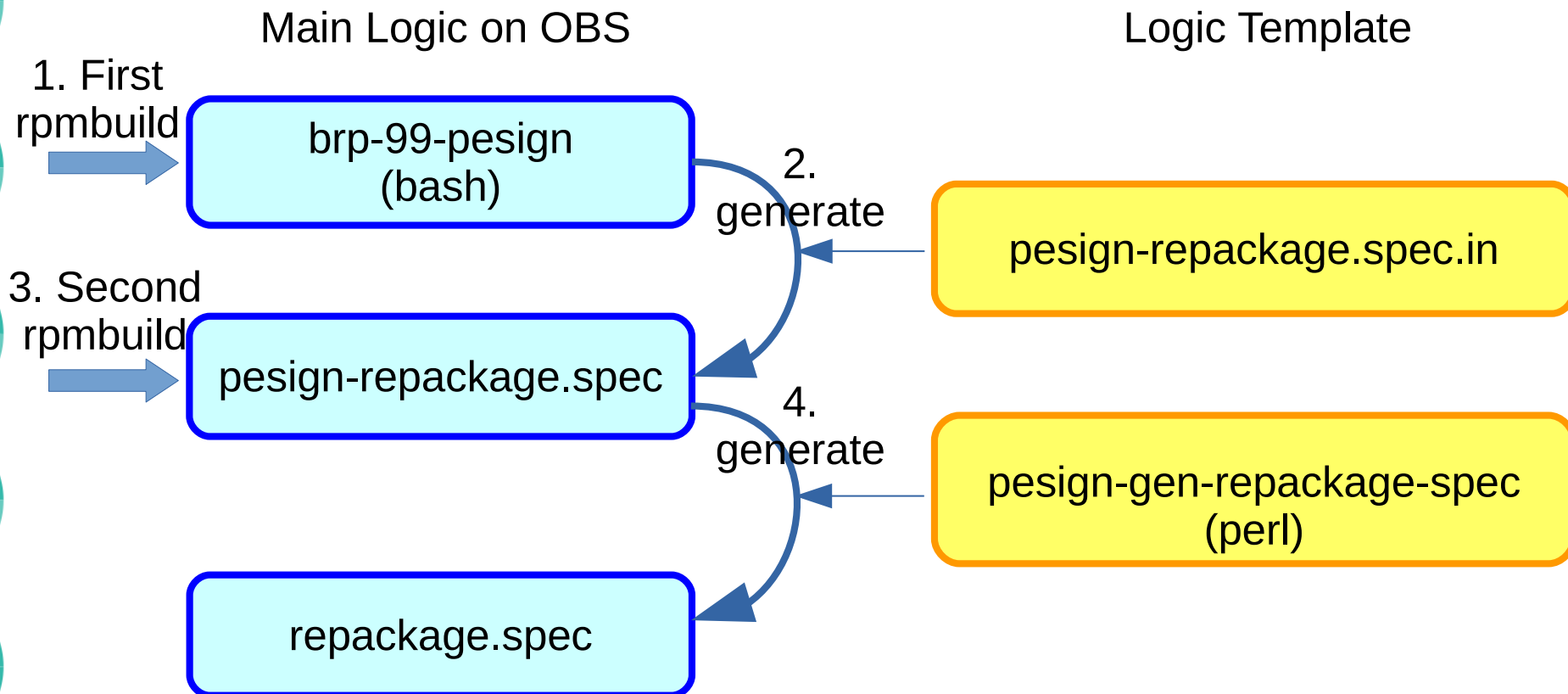
Execution flow



Execution flow



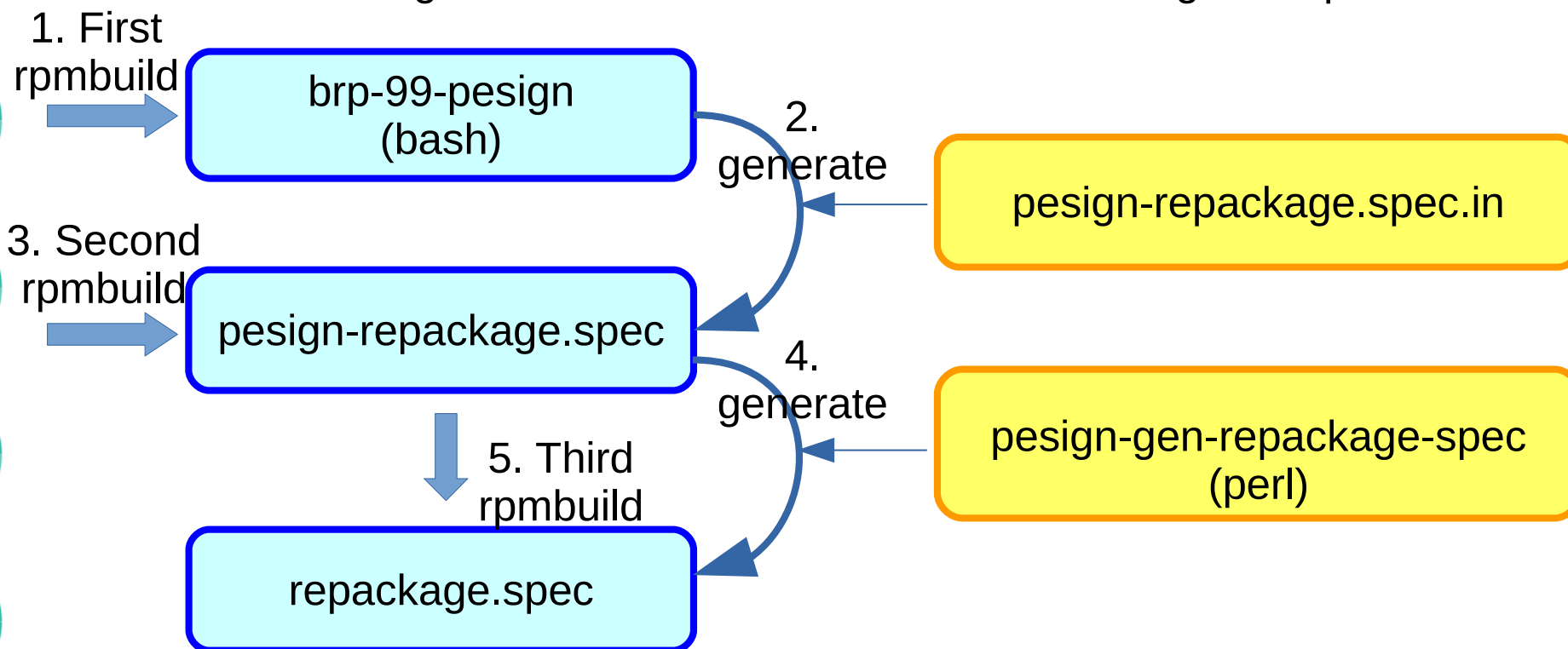
Execution flow



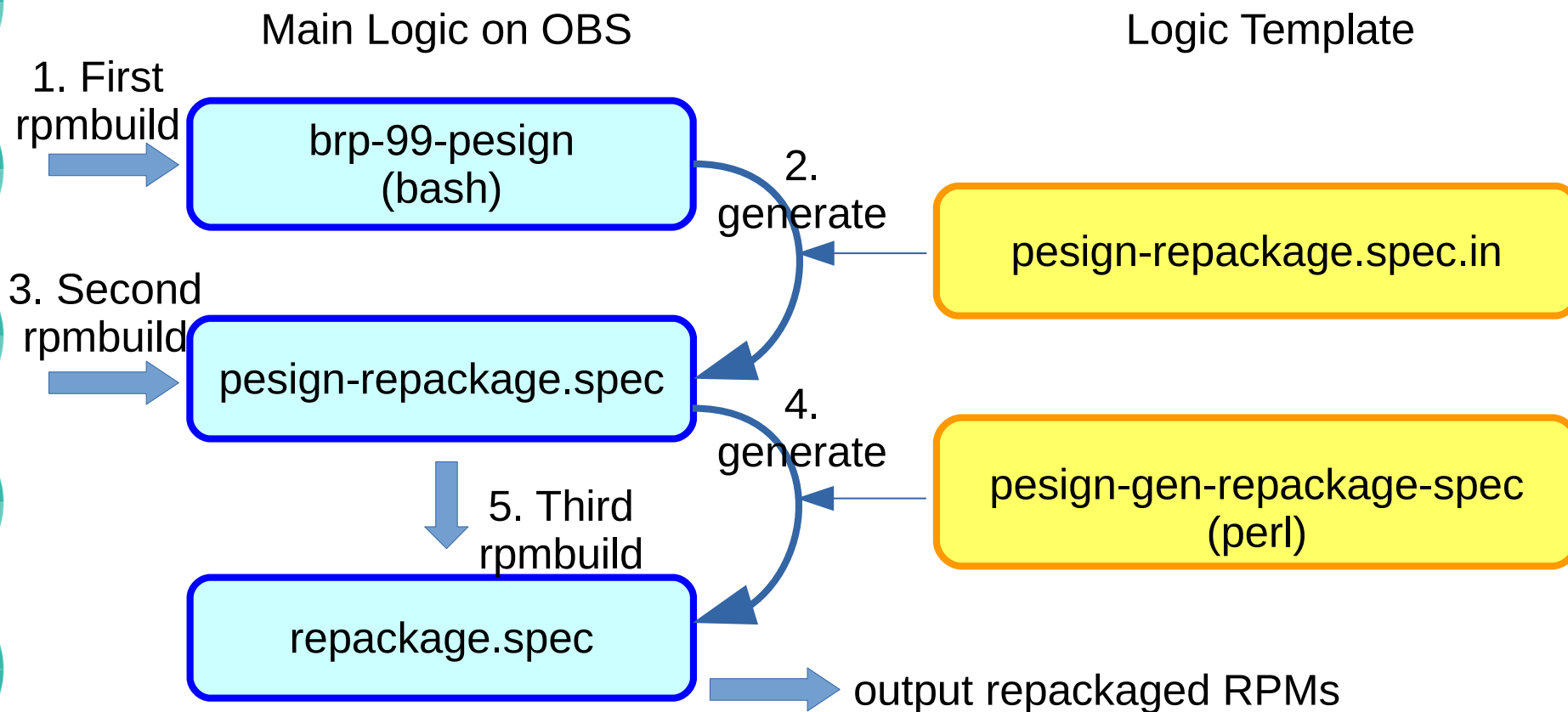
Execution flow

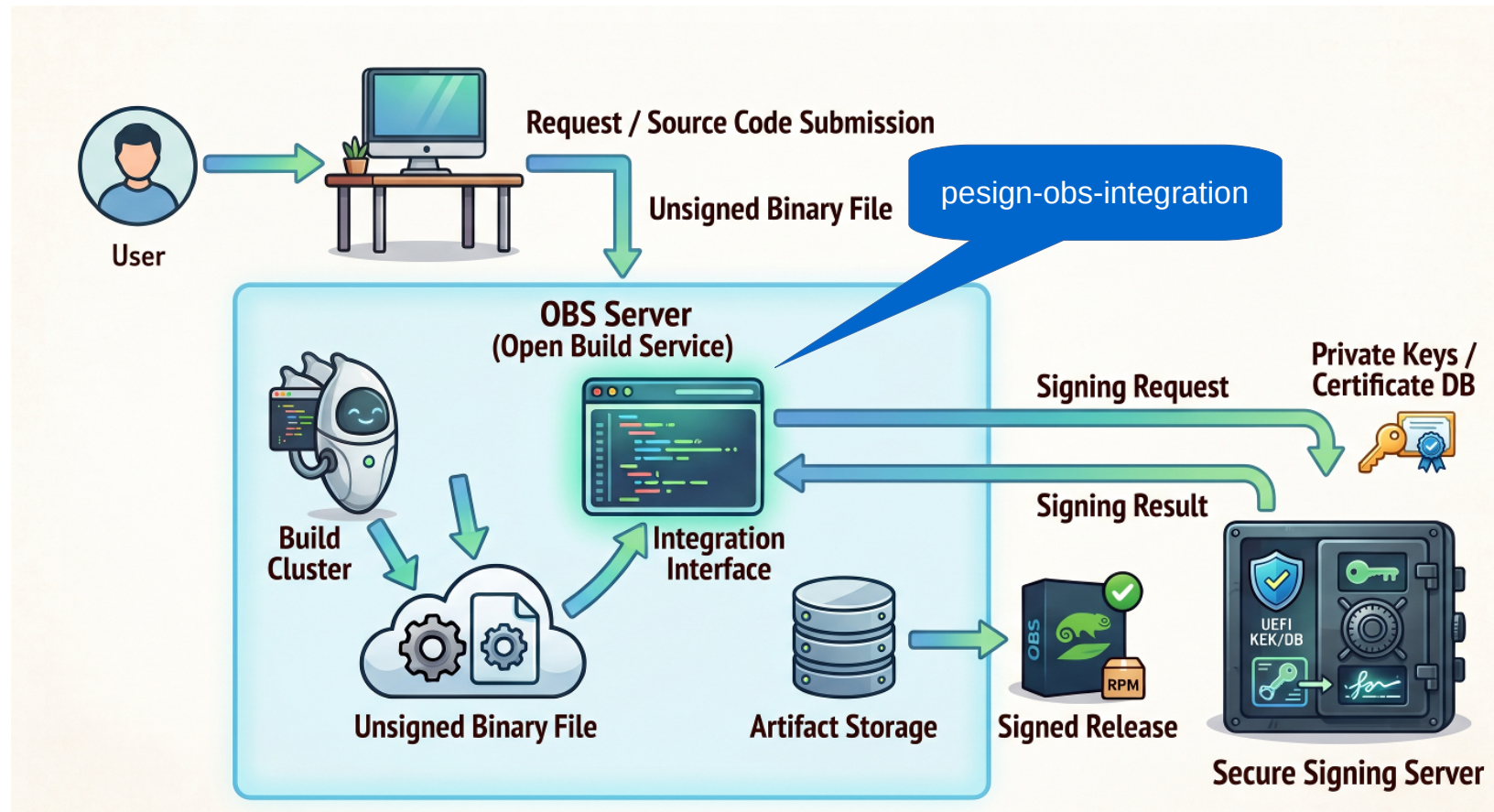
Main Logic on OBS

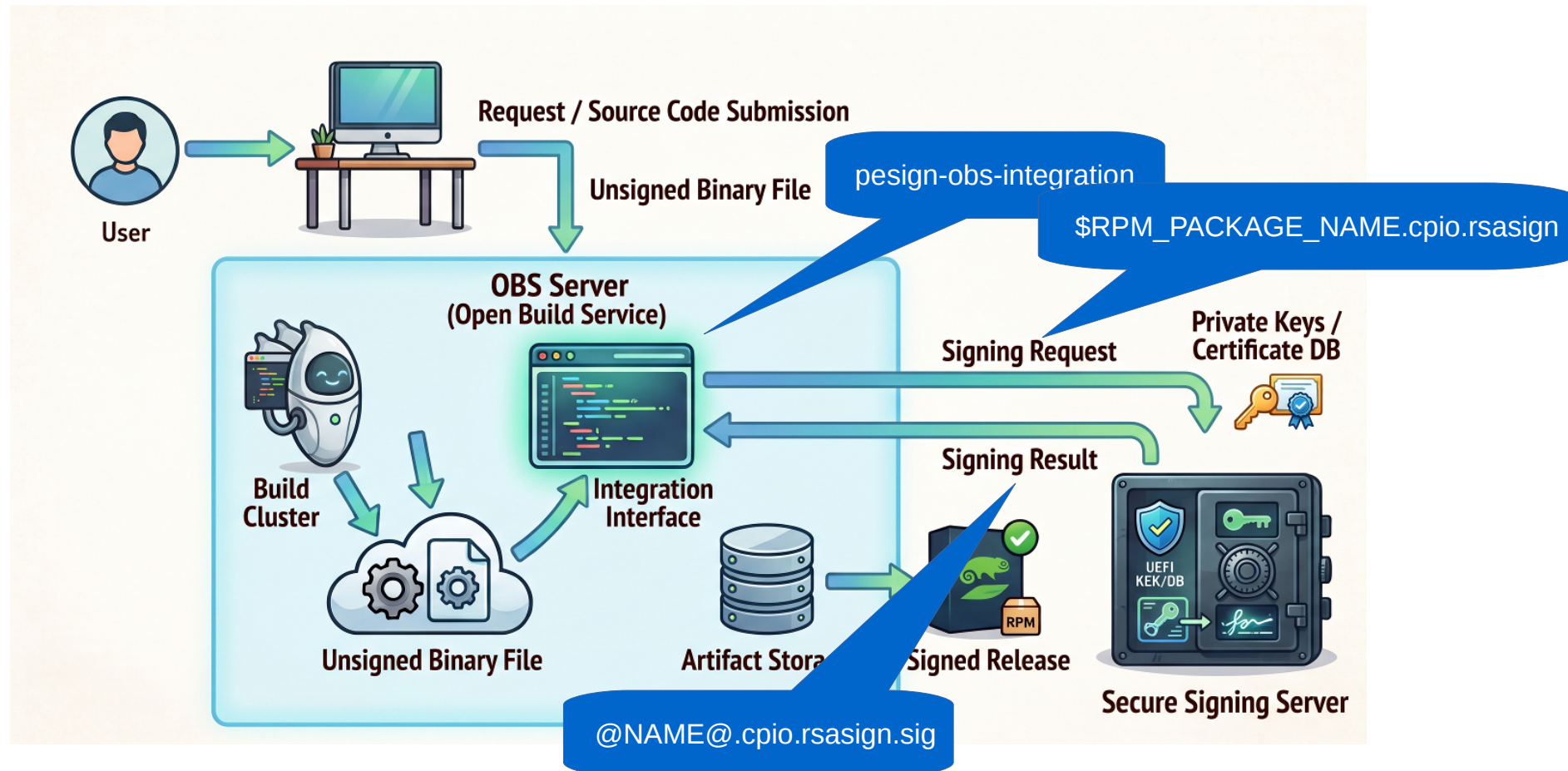
Logic Template



Execution flow







The flow for signing

Main Logic on OBS

1.
rpmbuild

brp-99-pesign
(bash)

pesign-repackage.spec

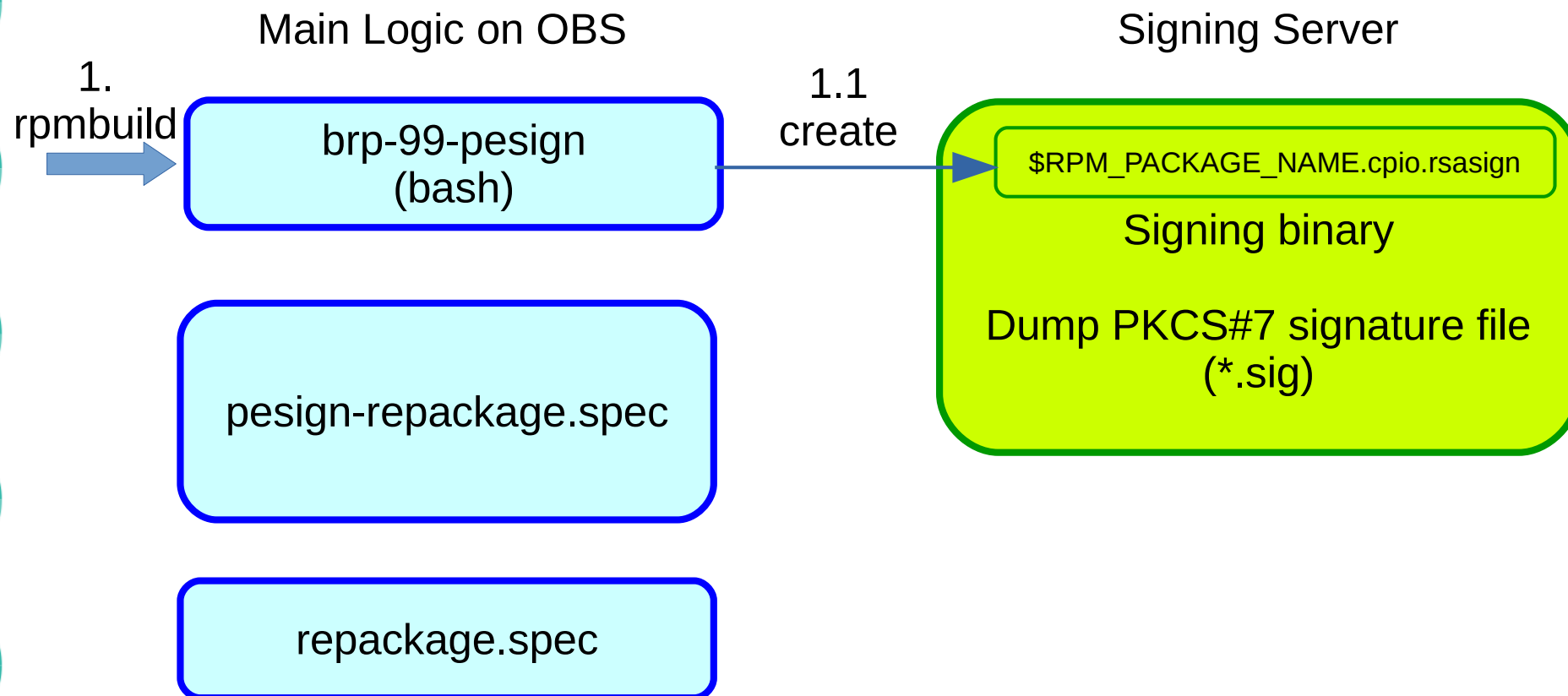
repackage.spec

Signing Server

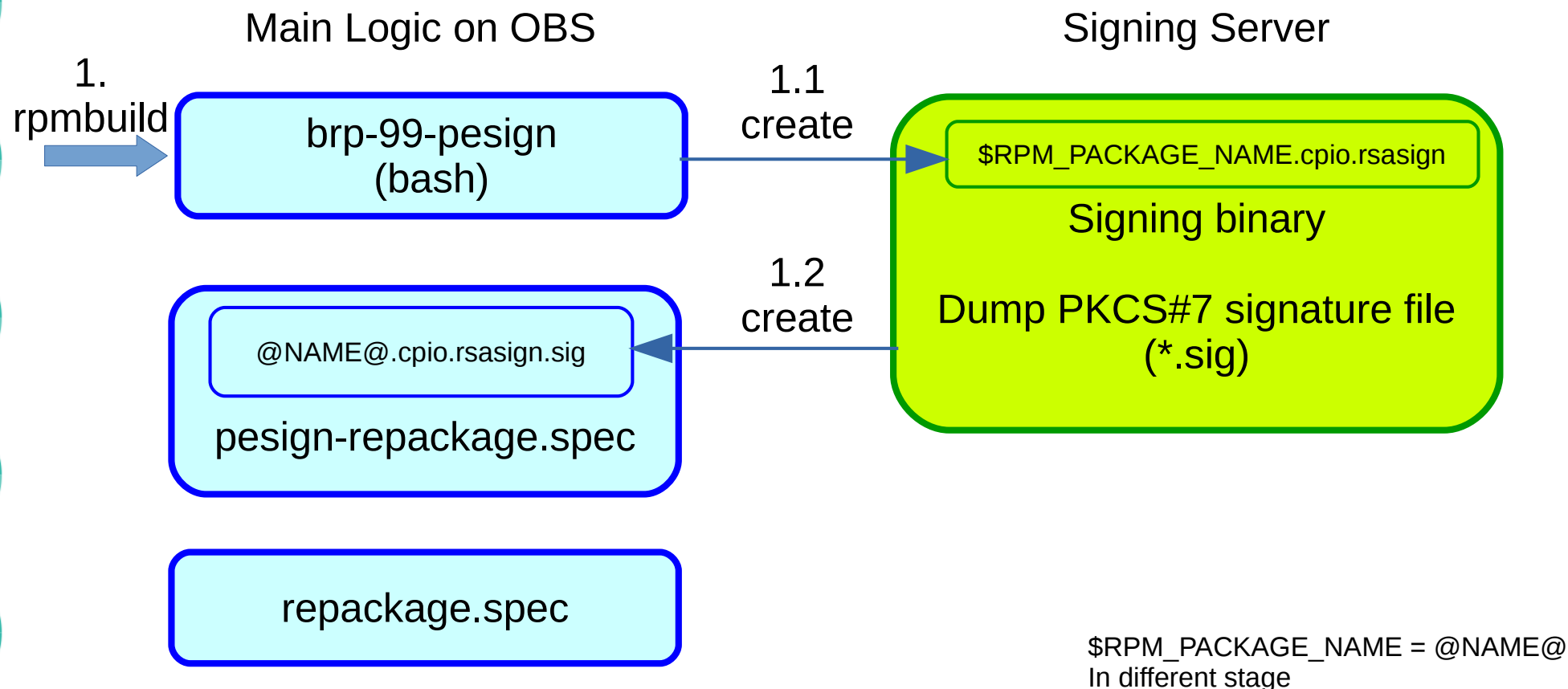
Signing binary

Dump PKCS#7 signature file
(* .sig)

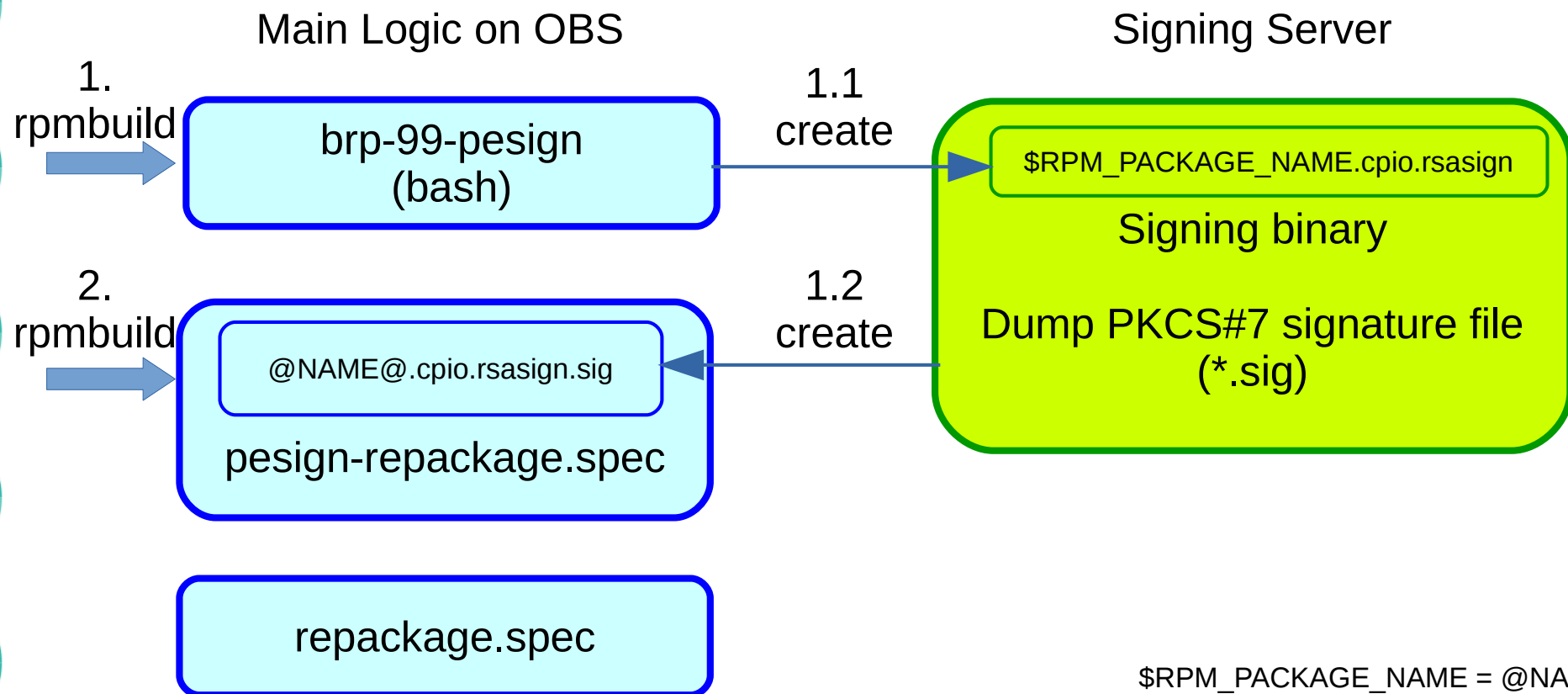
The flow for signing



The flow for signing



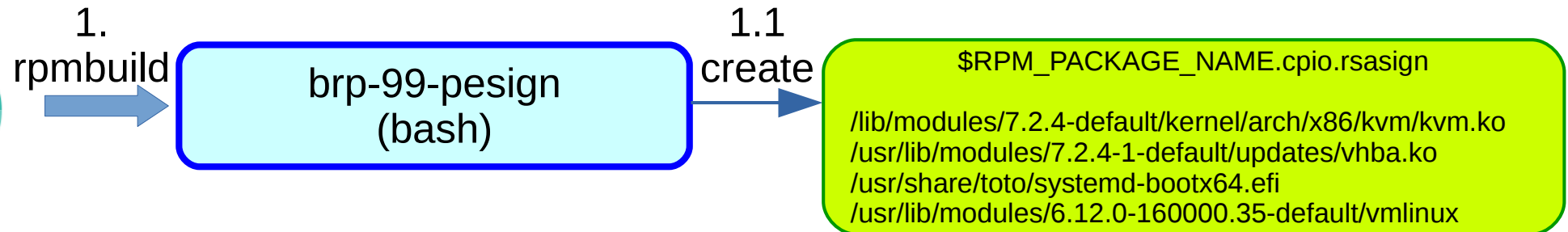
The flow for signing



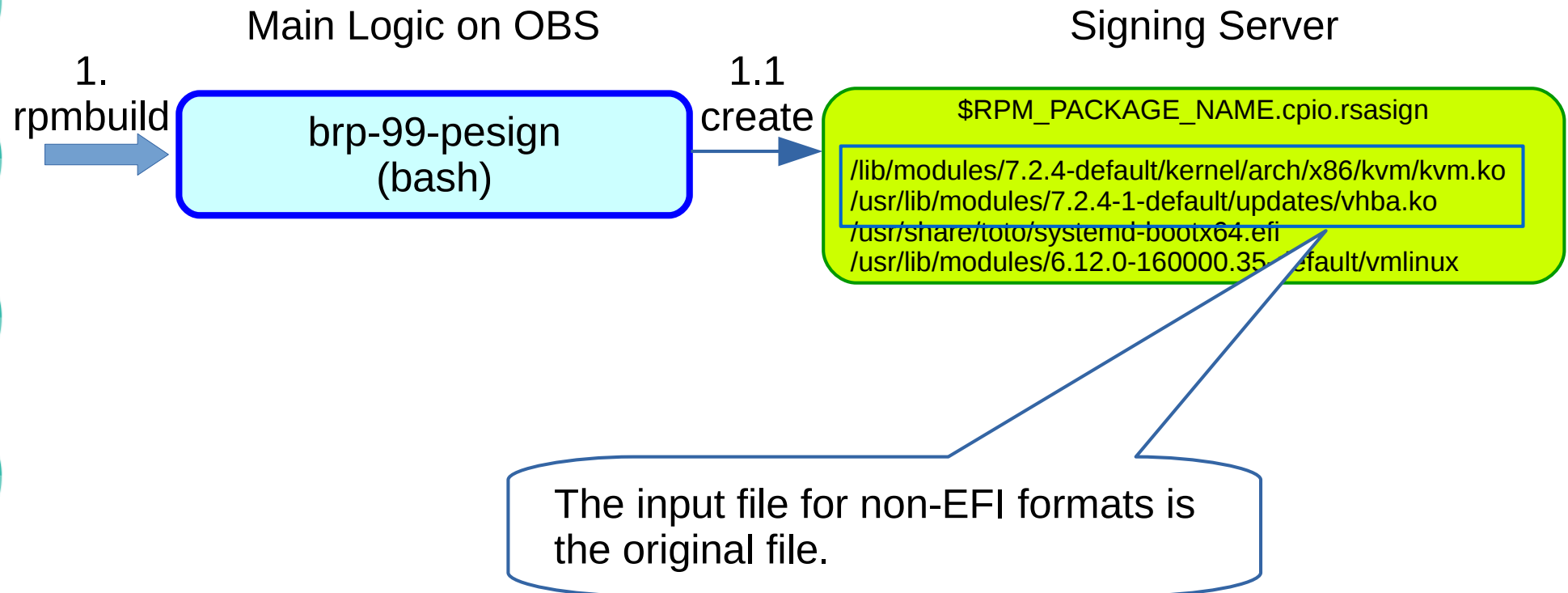
\$RPM_PACKAGE_NAME.cpio.rsasign

Main Logic on OBS

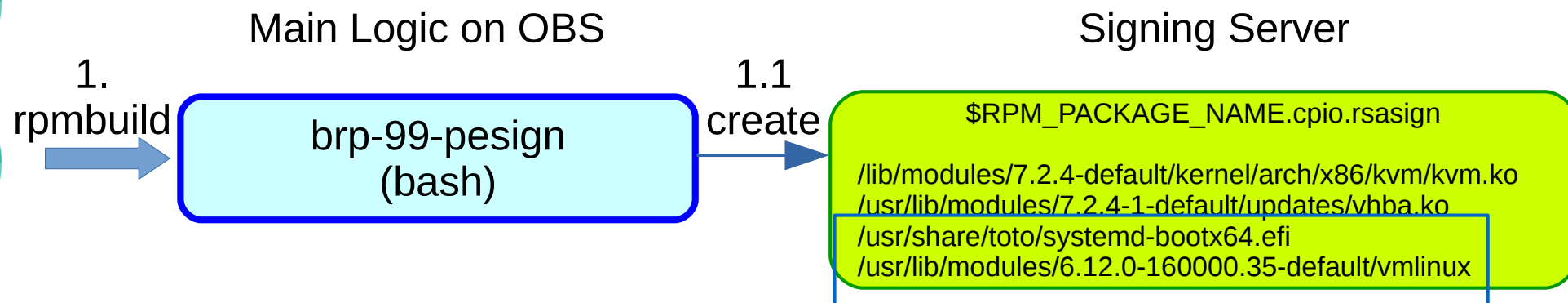
Signing Server



The input file of non-EFI binary



The input file of EFI binary



For EFI binaries, the input file for the signing Server is the PKCS#7 SignedAttributes.

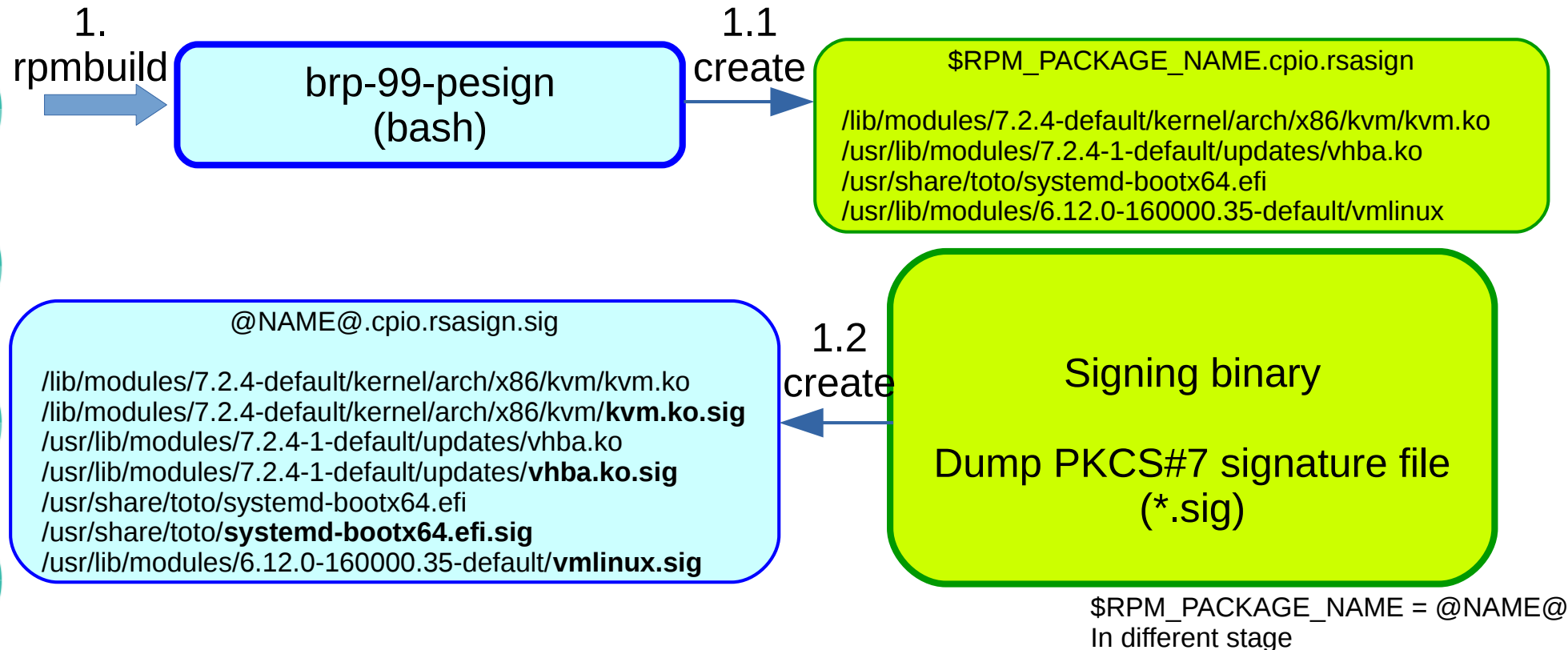
It's produced by:

pesign --certdir="\$nss_db" -i "\$f" -E \$dest

@NAME@.cpio.rsasign.sig

Main Logic on OBS

Signing Server



Steps in each stage

Main Logic on OBS

brp-99-pesign

1. Create \$RPM_PACKAGE_NAME.cpio.rsasign archive
2. Generate pesign-repackage.spec

Logic Template

psign-repackage.spec.in

psign-gen-repackage-spec
(perl)

Steps in each stage

Main Logic on OBS

brp-99-pesign

1. Create \$RPM_PACKAGE_NAME.cpio.rsasign archive
2. Generate pesign-repackage.spec

pesign-repackage.spec

1. Decompress RPMs
2. Unpack @NAME@.cpio.rsasign.sig
3. Verify the signature with the public key of the certificate
4. Attach the signature to target file
5. Preparing buildroot for repackaging stage
6. Generate repackage.spec
7. rpmbuild repackage.spec

generate

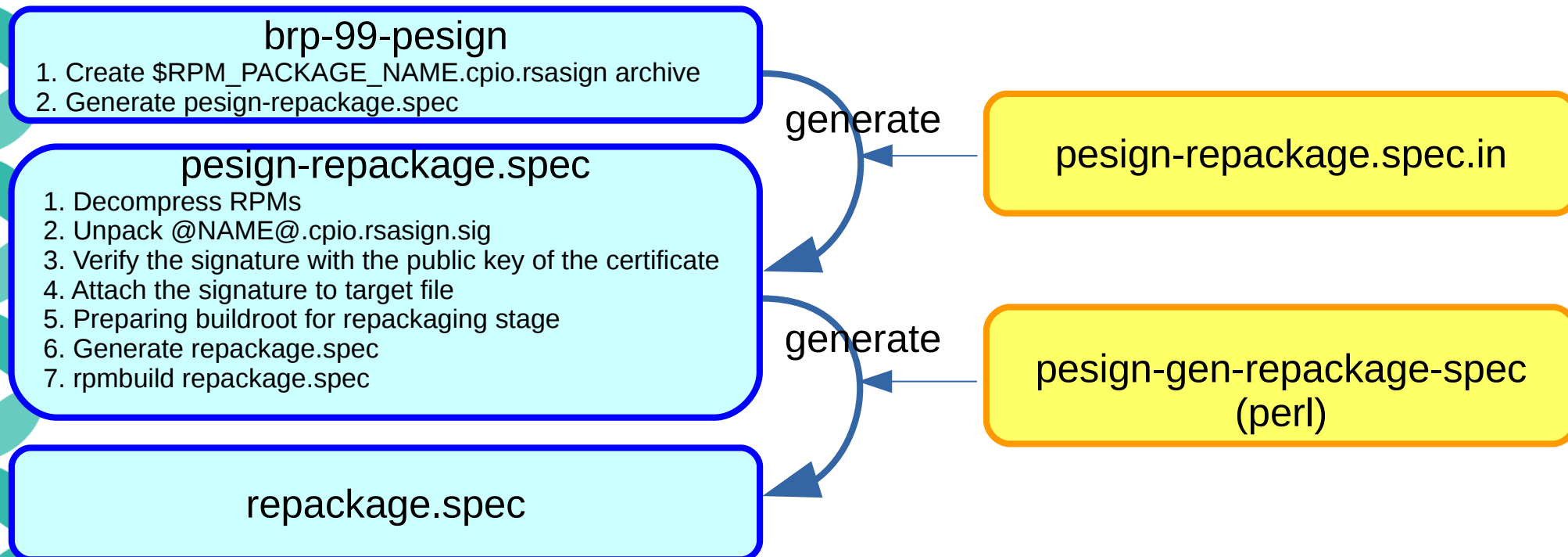
pesign-repackage.spec.in

pesign-gen-repackage-spec
(perl)

Steps in each stage

Main Logic on OBS

Logic Template



Steps in each stage

Main Logic on OBS

Stage 1. Creating *.cpio.rsasign archive

brp-99-pesign

1. Create \$RPM_PACKAGE_NAME.cpio.rsasign archive
2. Generate pesign-repackage.spec

pesign-repackage.spec

1. Decompress RPMs
2. Unpack @NAME@.cpio.rsasign.sig
3. Verify the signature with the public key of the certificate
4. Attach the signature to target file
5. Preparing buildroot for repackaging stage
6. Generate repackage.spec
7. rpmbuild repackage.spec

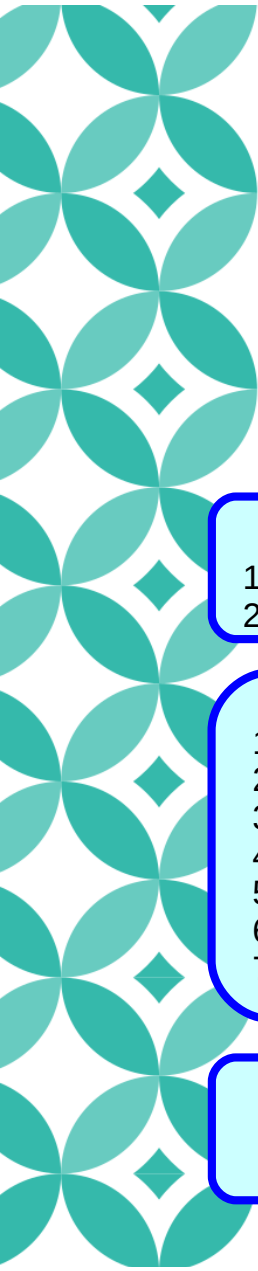
repackage.spec

pesign-repackage.spec.in

pesign-gen-repackage-spec
(perl)

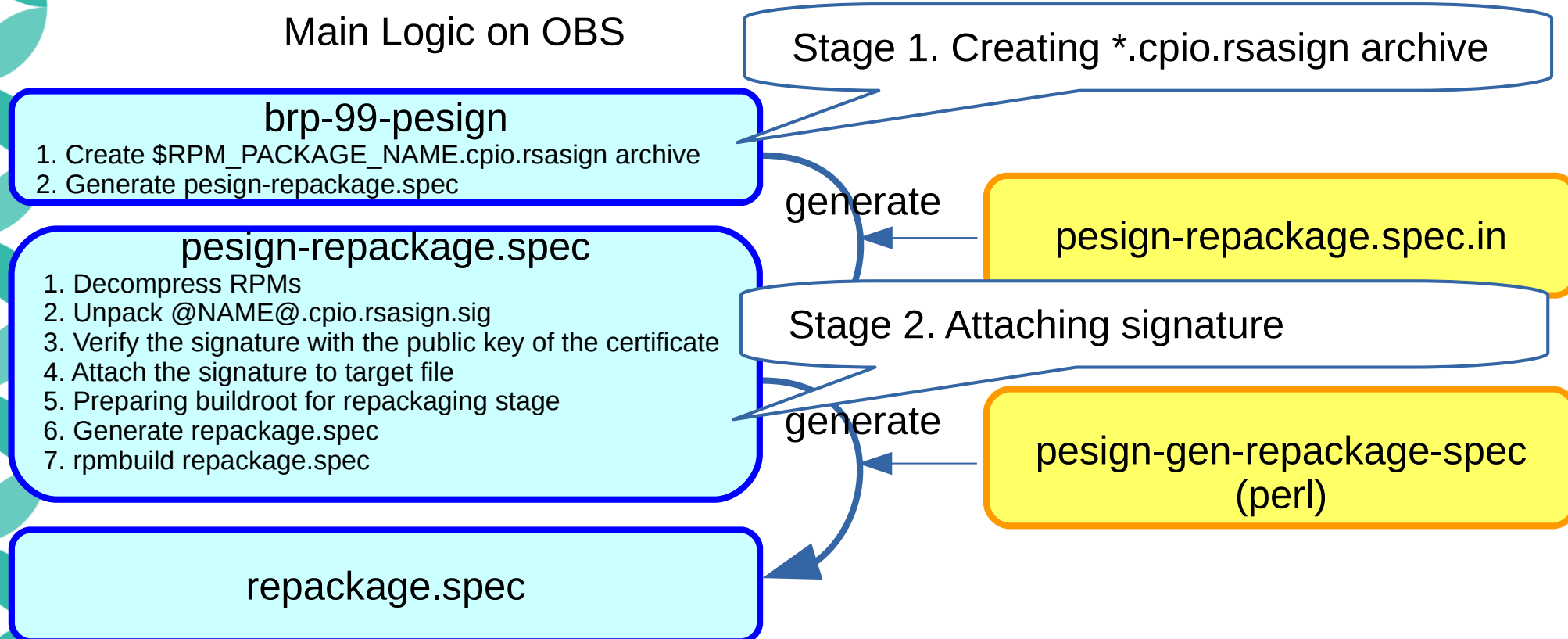
generate

generate



Steps in each stage

Main Logic on OBS



Steps in each stage

Main Logic on OBS

brp-99-pesign

1. Create \$RPM_PACKAGE_NAME.cpio.rsasign archive
2. Generate pesign-repackage.spec

Stage 1. Creating *.cpio.rsasign archive

pesign-repackage.spec

1. Decompress RPMs
2. Unpack @NAME@.cpio.rsasign.sig
3. Verify the signature with the public key of the certificate
4. Attach the signature to target file
5. Preparing buildroot for repackaging stage
6. Generate repackage.spec
7. rpmbuild repackage.spec

generate

pesign-repackage.spec.in

Stage 2. Attaching signature

generate

pesign-gen-repackage-spec
(perl)

Stage 3. Repackaging

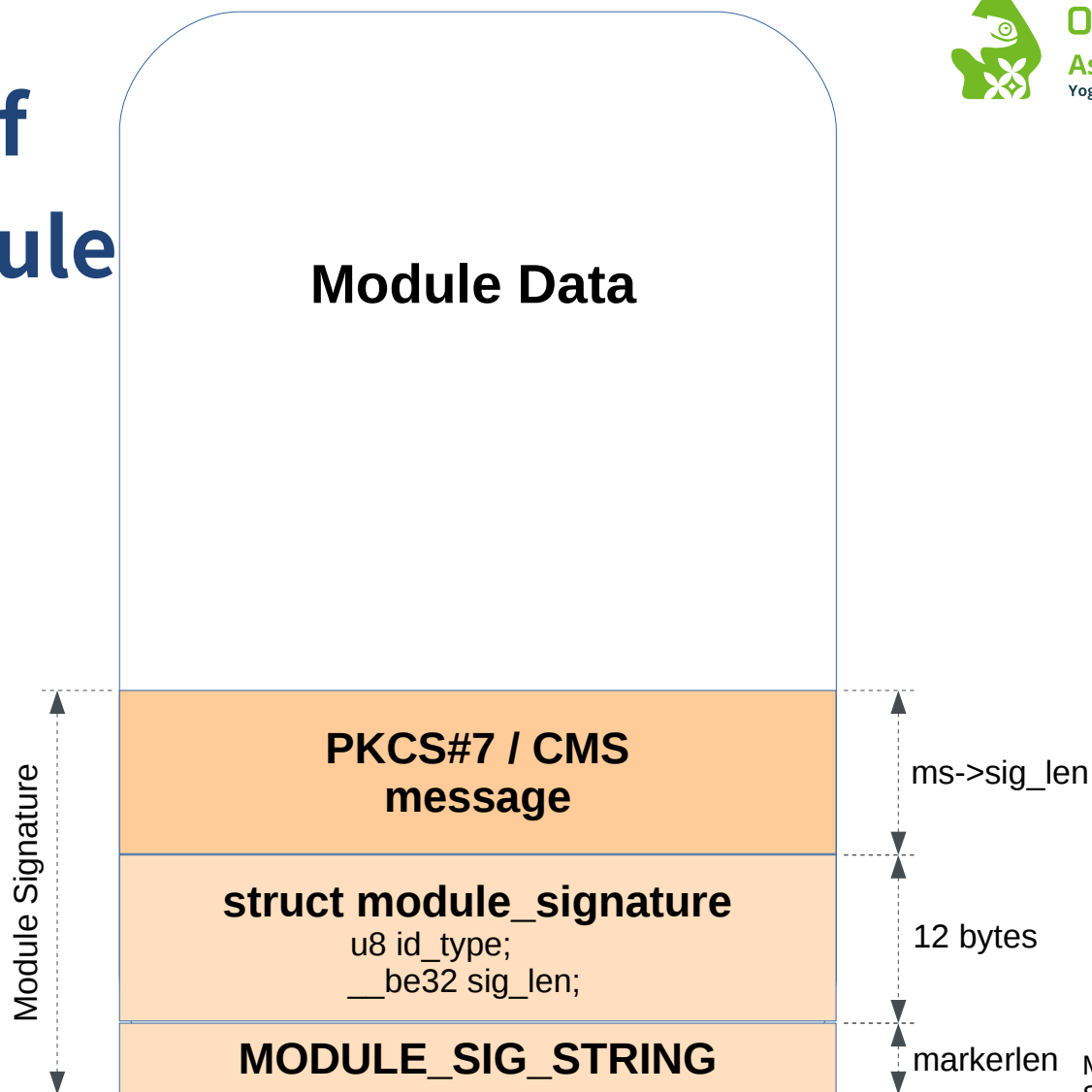
repackage.spec



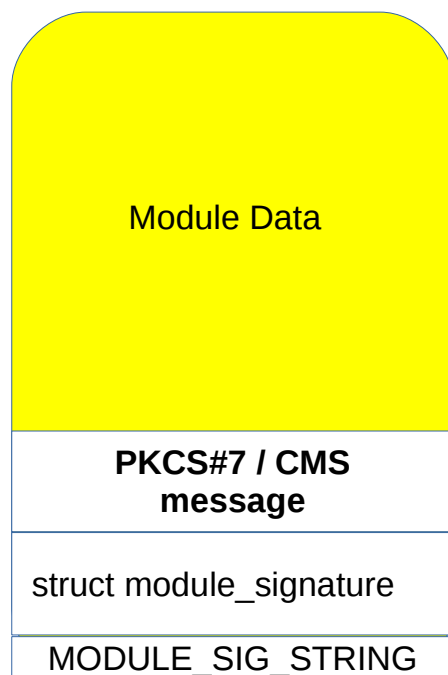
openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

Kernel module (ko) file signing

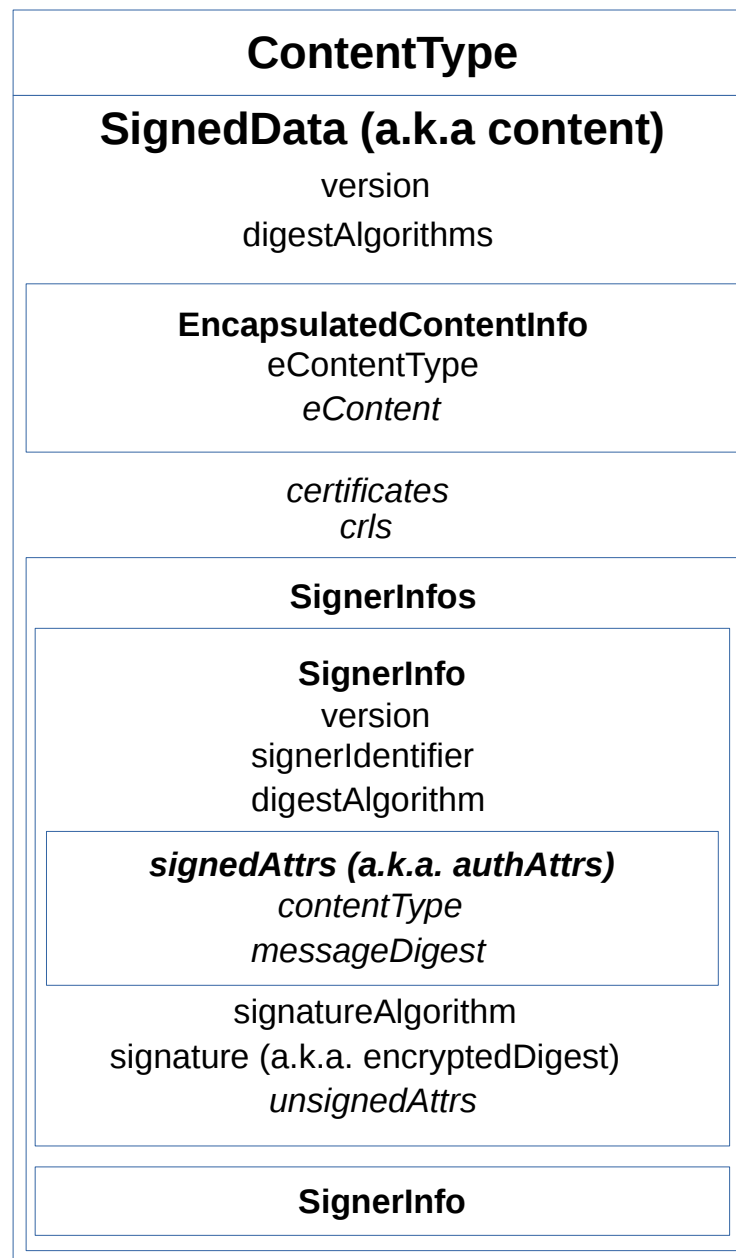
Signature of Kernel Module



CMS Message



Yellow regions: Data hashed to generate the unsigned digest.

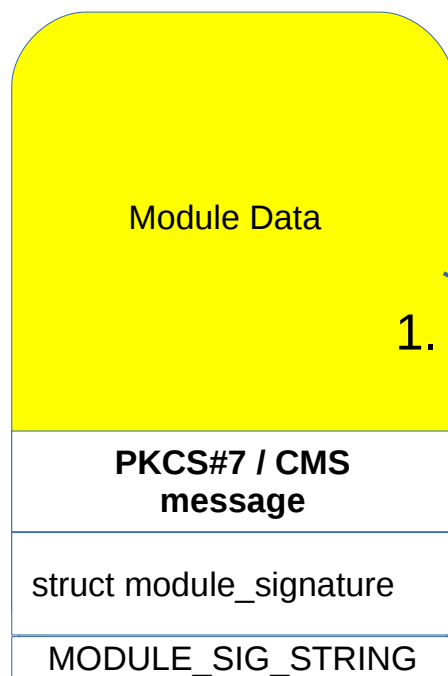


openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

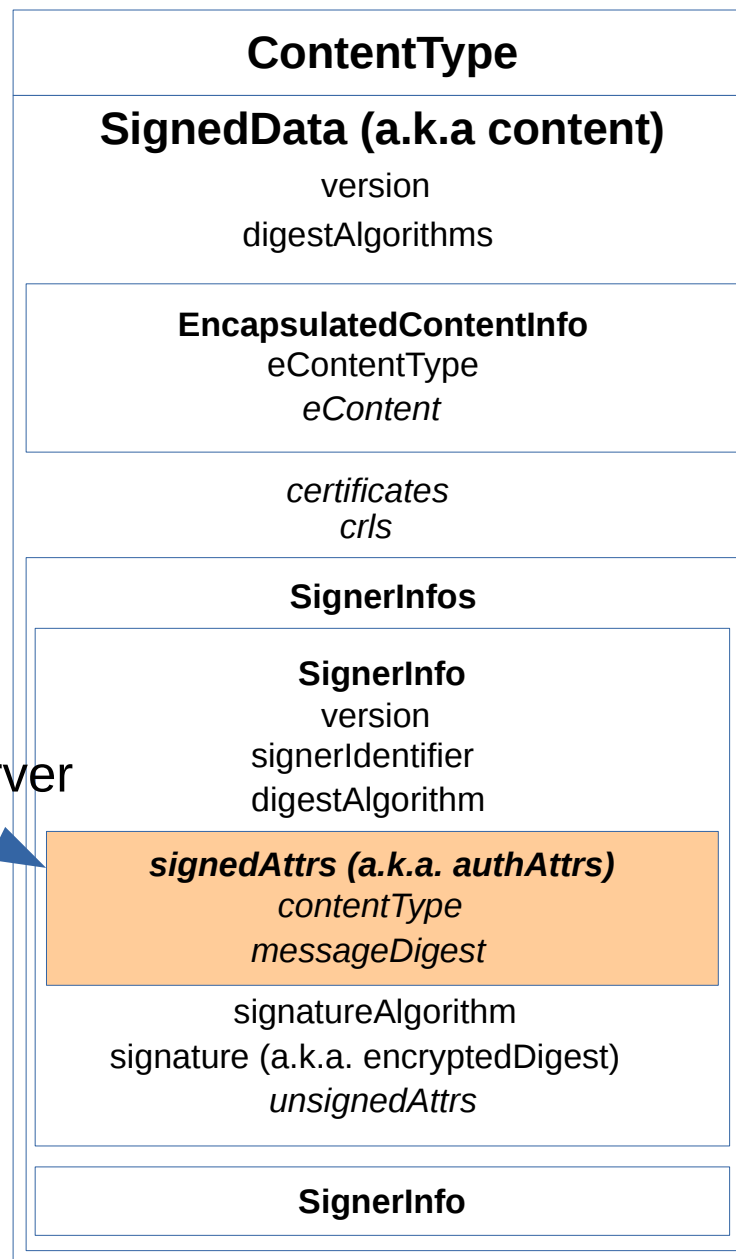
Multi-signed Kernel Module
SUSE Labs Conference 2020

CMS Message



Yellow regions: Data hashed to generate the unsigned digest.

1. Signing server

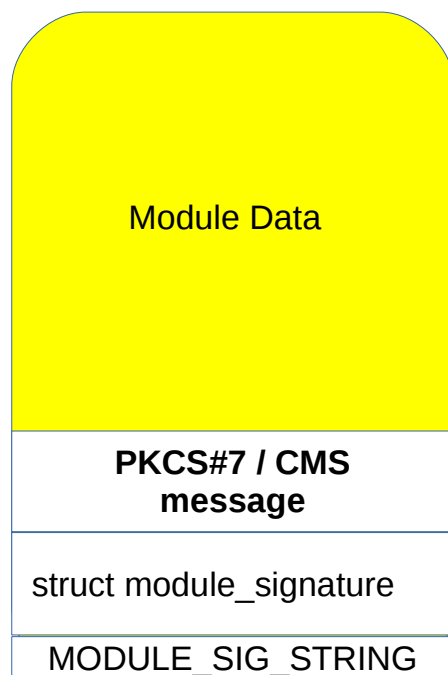


openSUSE[®]
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

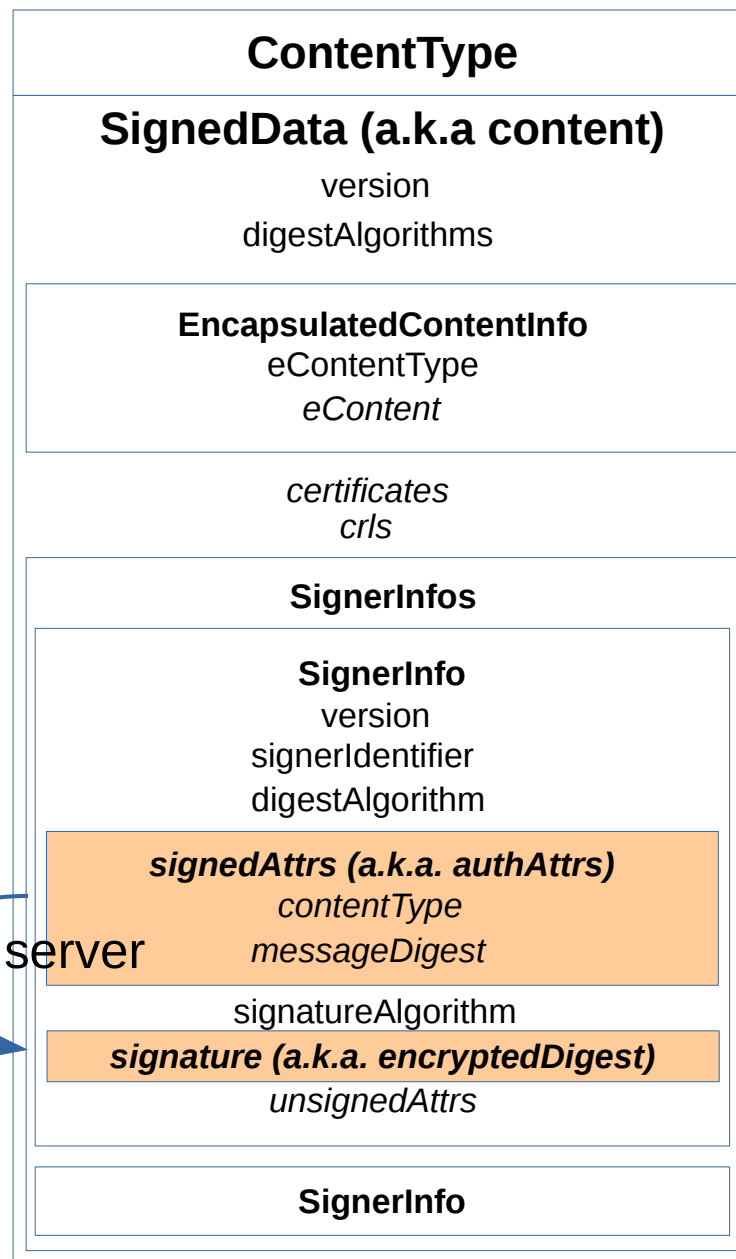
Multi-signed Kernel Module
SUSE Labs Conference 2020

CMS Message



Yellow regions: Data hashed to generate the unsigned digest.

2. Signing server



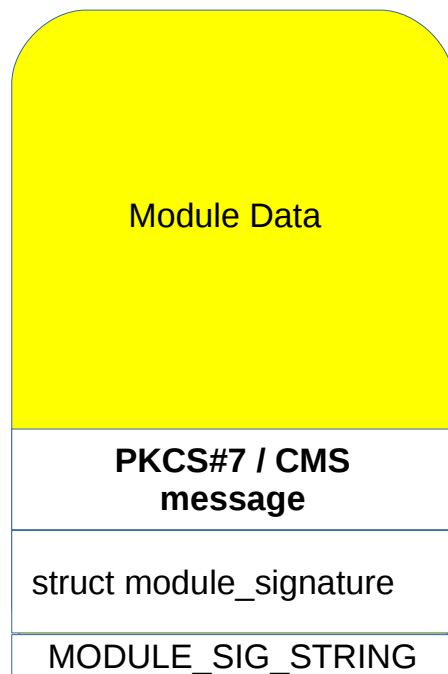
openSUSE[®]
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

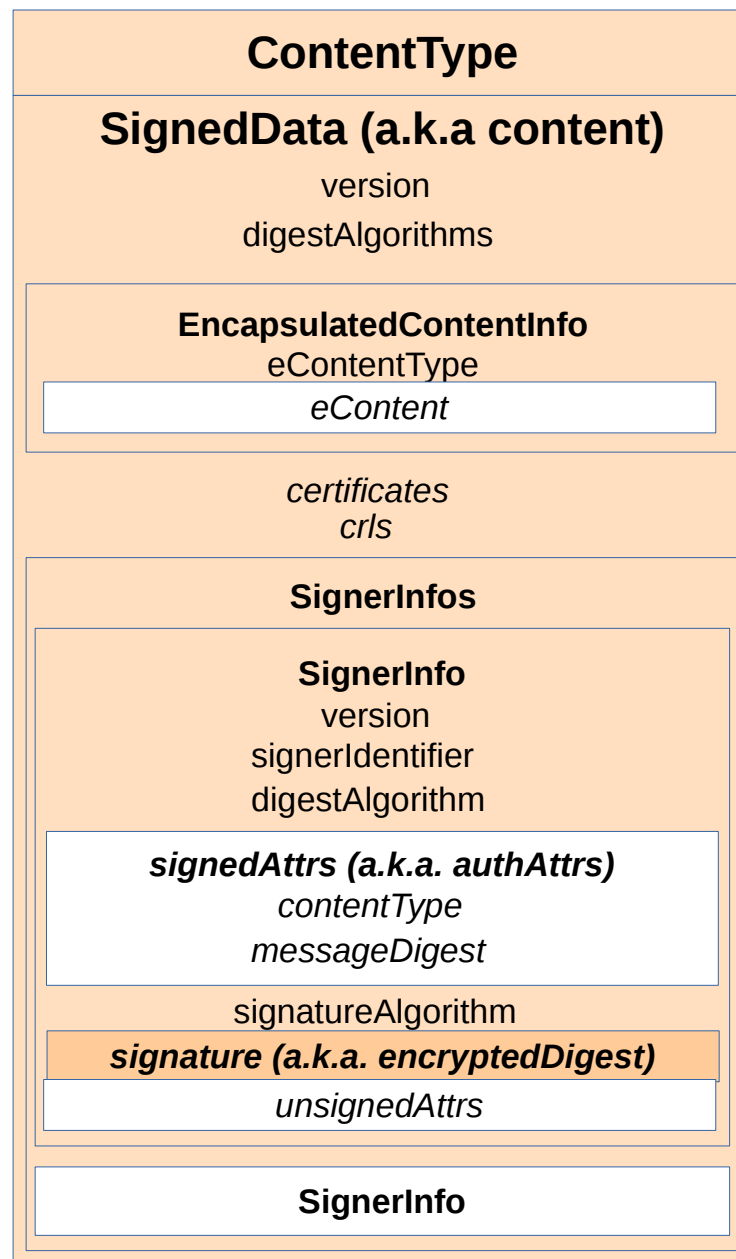
Multi-signed Kernel Module
SUSE Labs Conference 2020

CMS Message

3. Signing server
ASN.1 DER Encoding



Yellow regions: Data hashed to generate the unsigned digest.



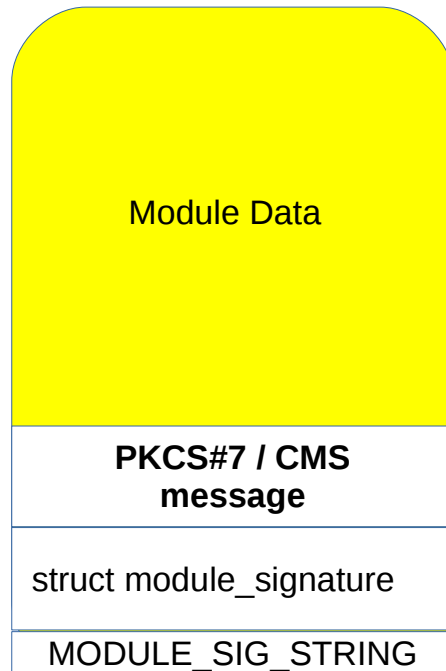
openSUSE[®]
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

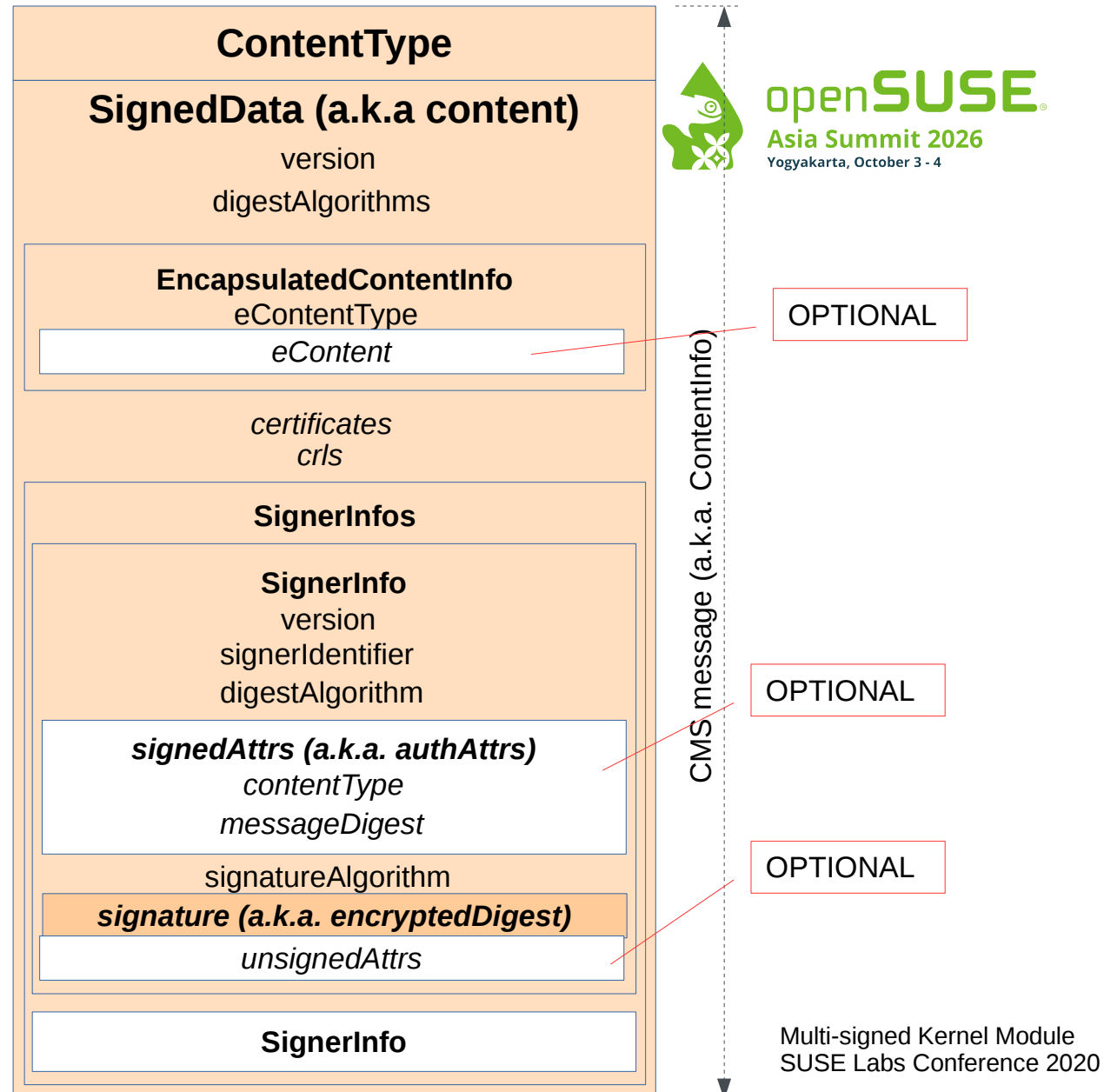
Multi-signed Kernel Module
SUSE Labs Conference 2020

CMS Message

3. Signing server
ASN.1 DER Encoding



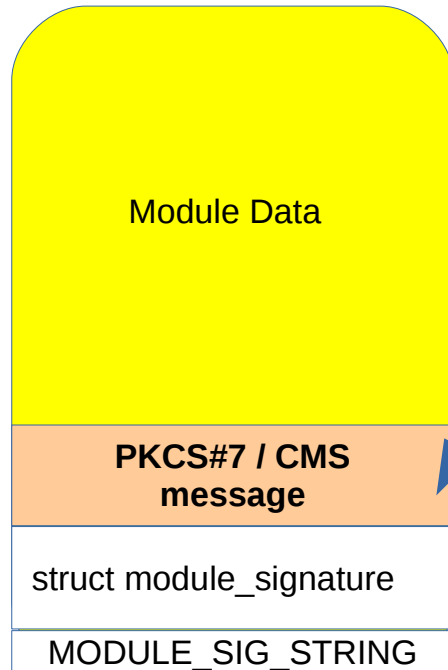
Yellow regions: Data hashed to generate the unsigned digest.



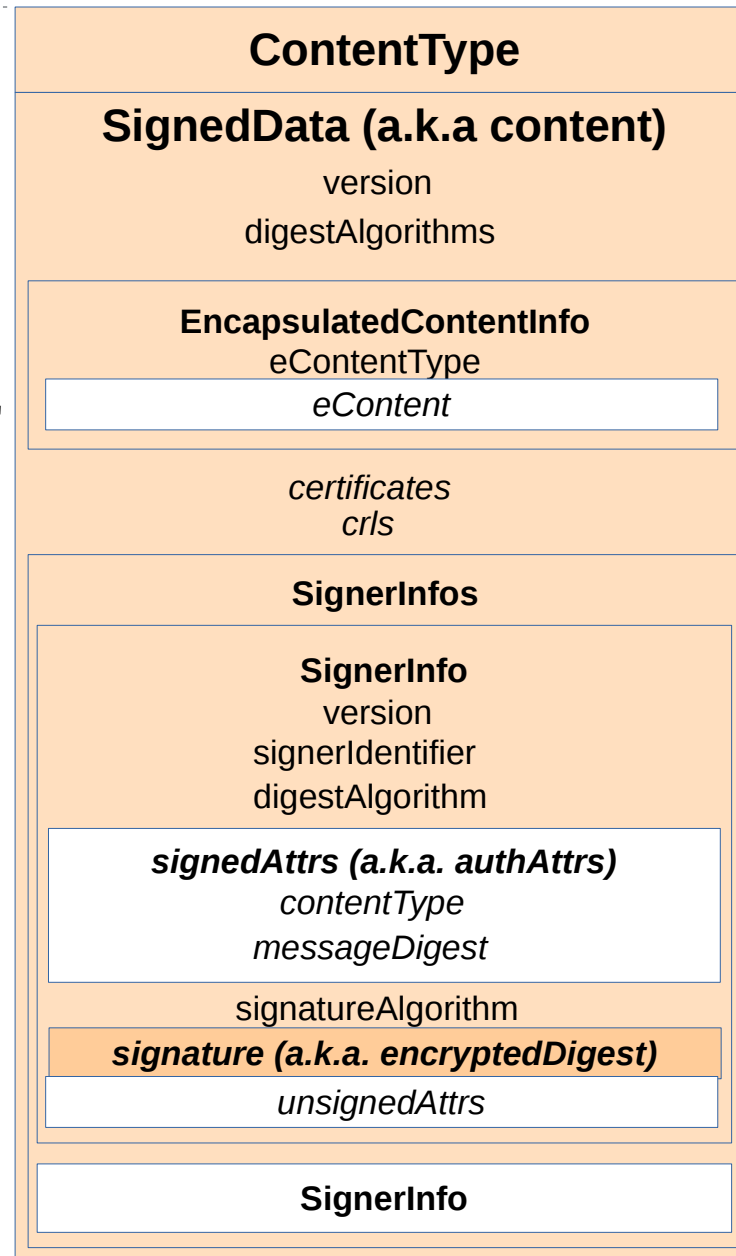
openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS Message

4. kernel-sign-file in
pesign-repackage.spec
-i pkcs7 -s "\$sig" sha256 "\$cert" "\$f"



Yellow regions: Data hashed to generate the unsigned digest.



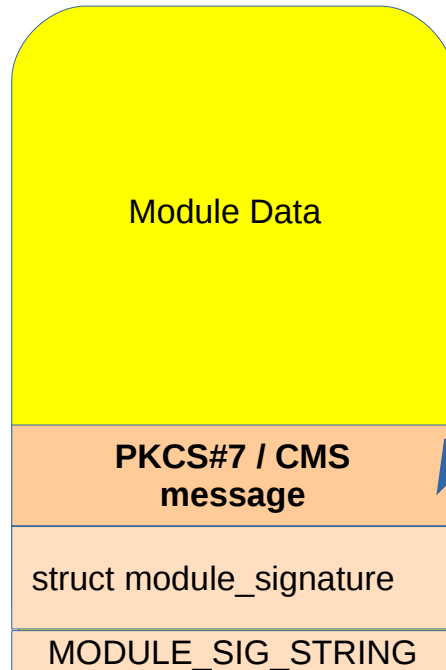
openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

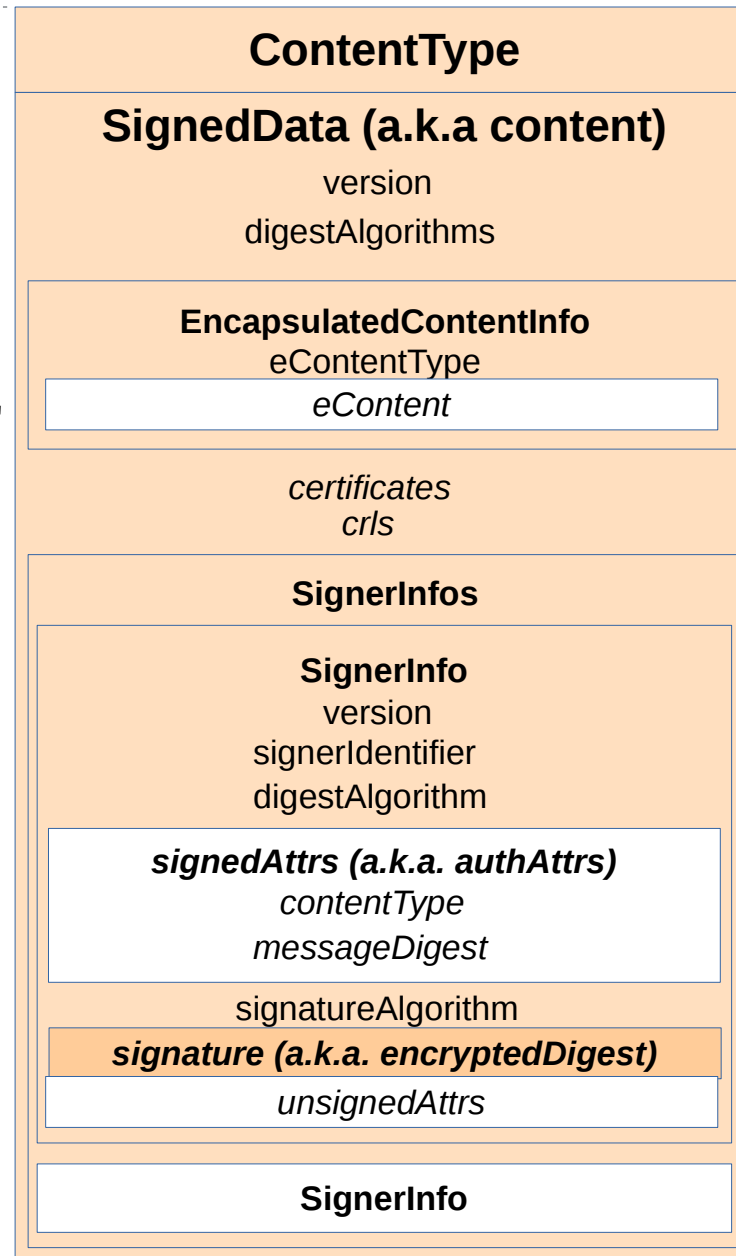
Multi-signed Kernel Module
SUSE Labs Conference 2020

CMS Message

4. kernel-sign-file in
pesign-repackage.spec
-i pkcs7 -s "\$sig" sha256 "\$cert" "\$f"



Yellow regions: Data hashed to
generate the unsigned digest.



openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

CMS message (a.k.a. ContentInfo)

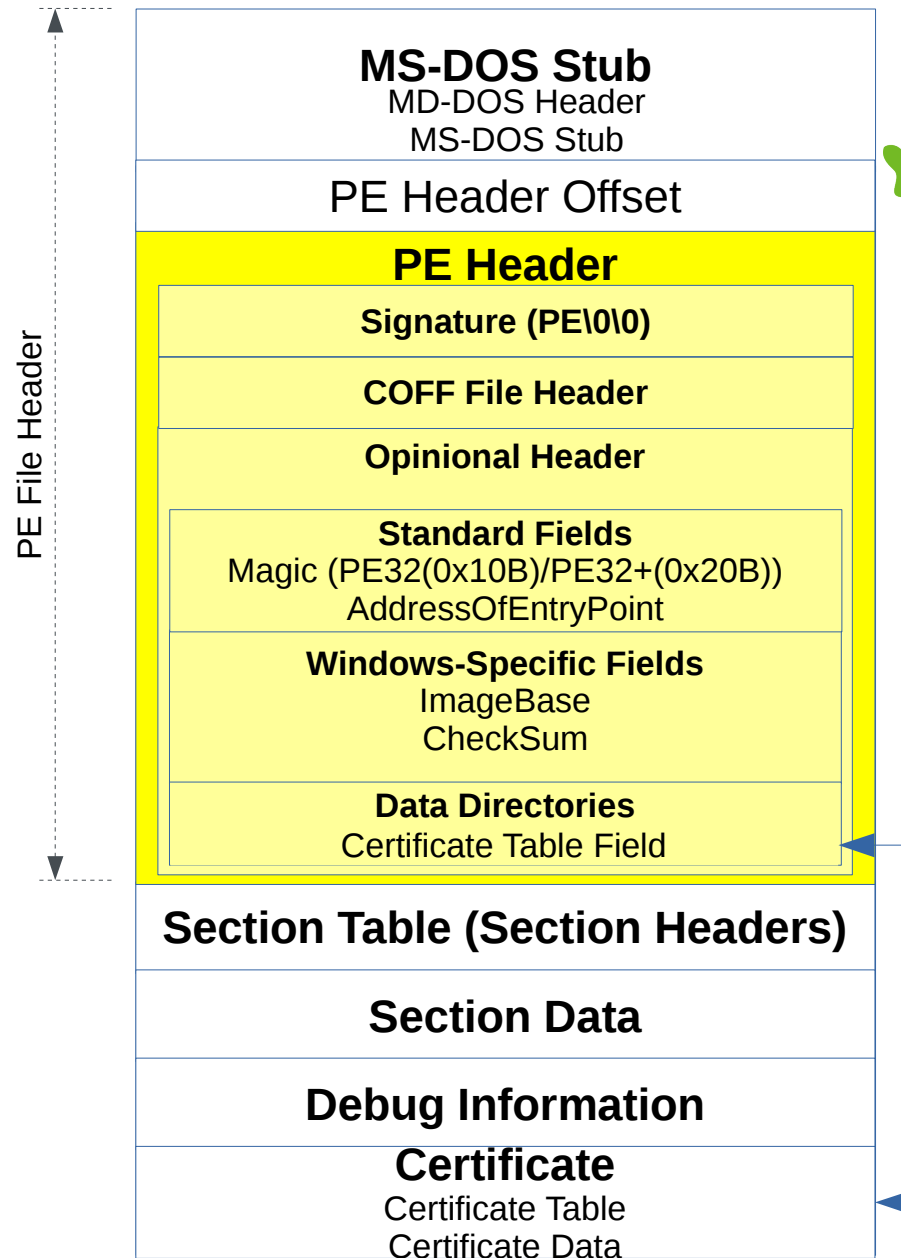
Multi-signed Kernel Module
SUSE Labs Conference 2020



openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

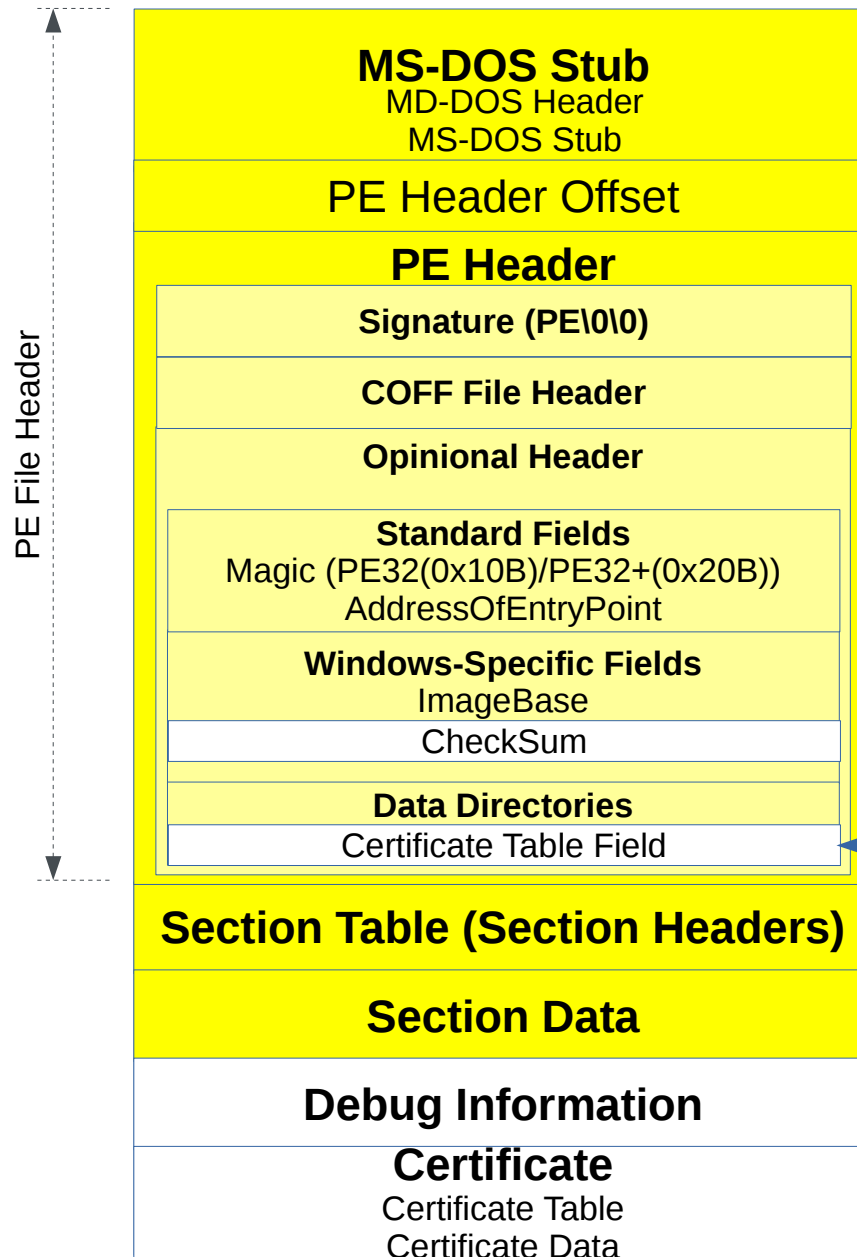
PE/COFF (efi) binary file signing

PE/COFF (.efi) File Structure



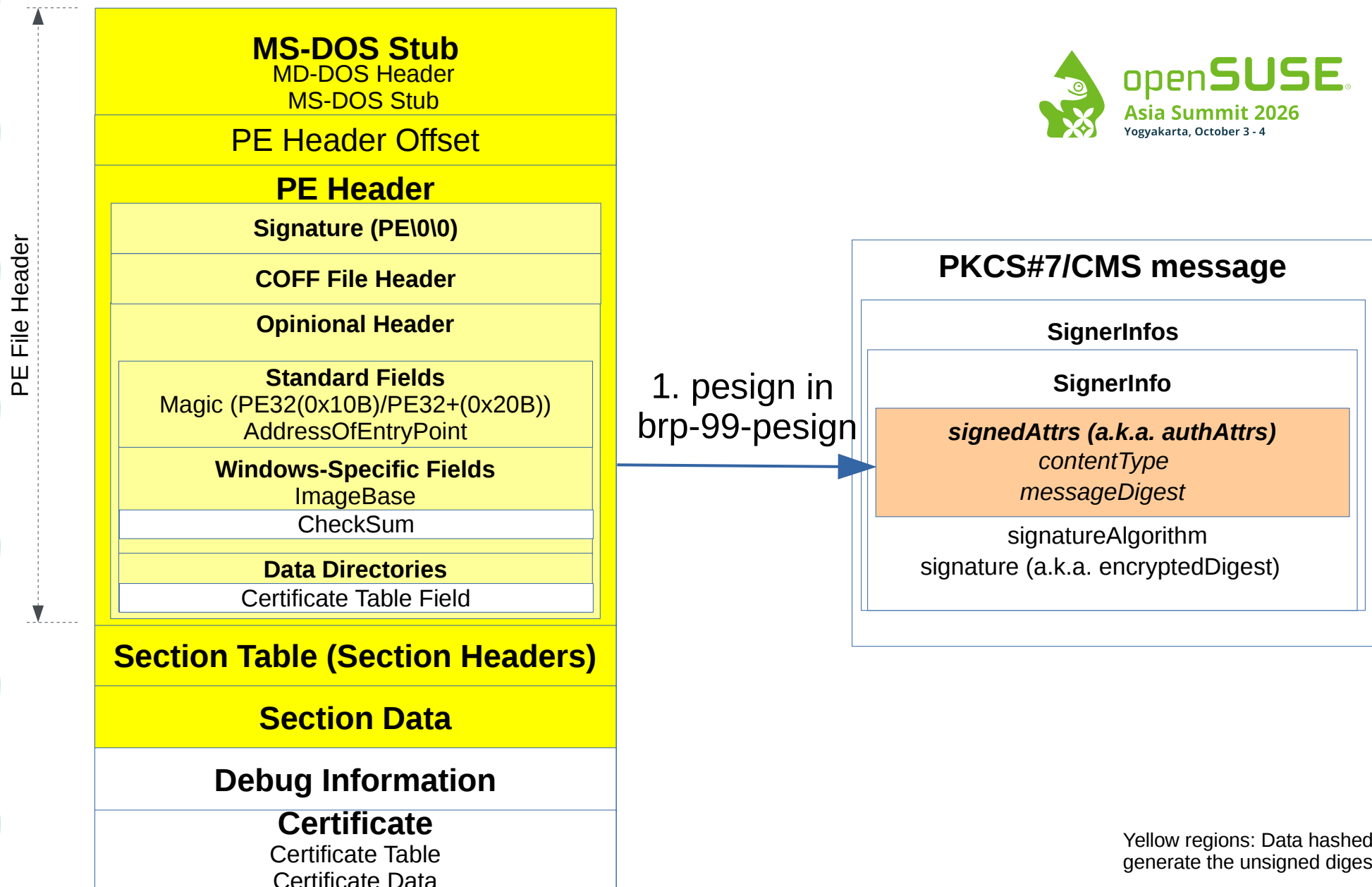
openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

PE/COFF (.efi) File Hashing Range



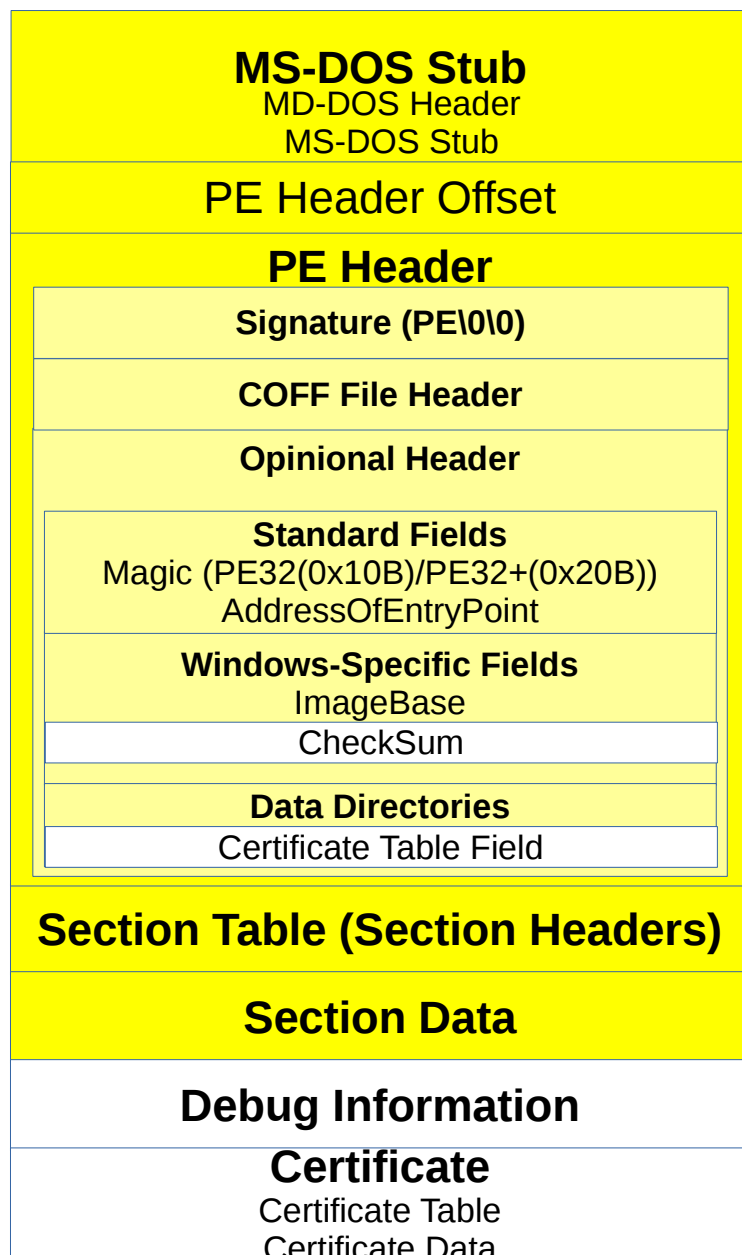
openSUSE®
Asia Summit 2026
Yogyakarta, October 3 - 4

Yellow regions: Data hashed to generate the unsigned digest.

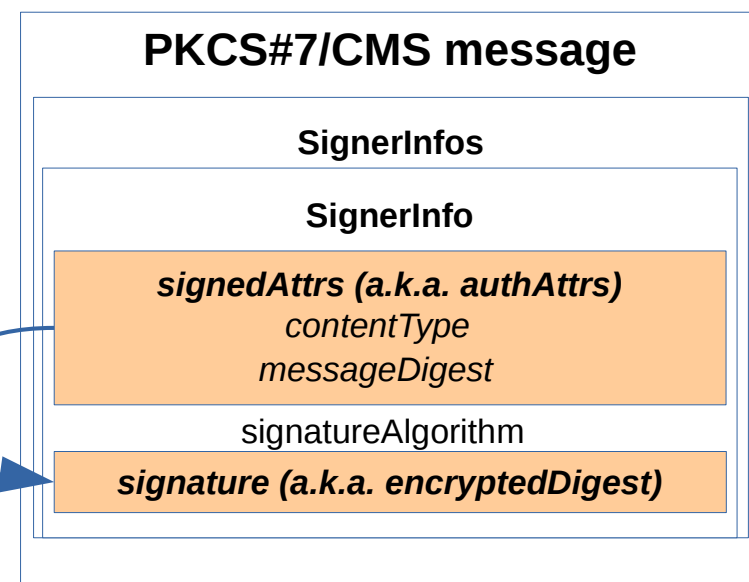




PE File Header



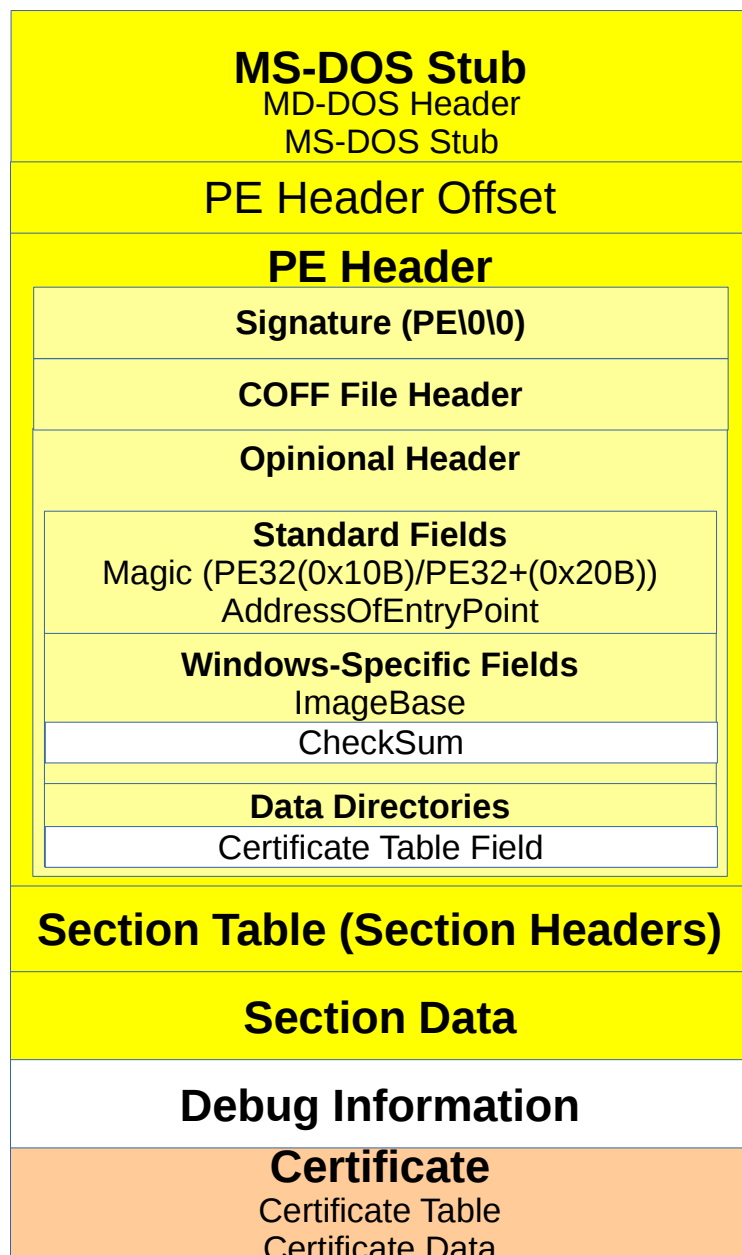
2. Signing server



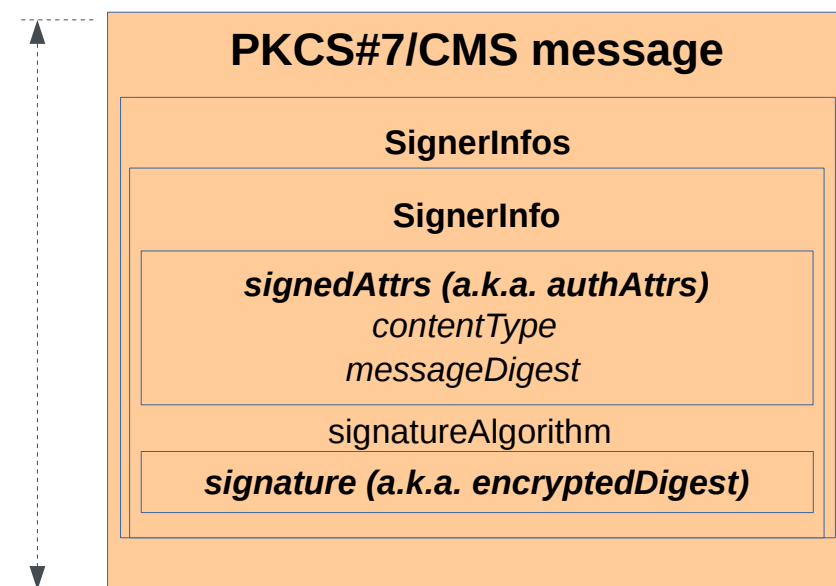
Yellow regions: Data hashed to generate the unsigned digest.



PE File Header



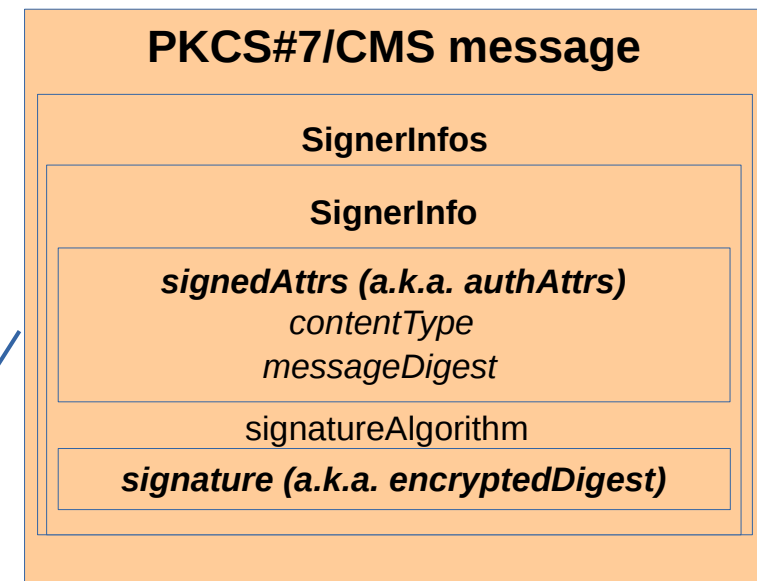
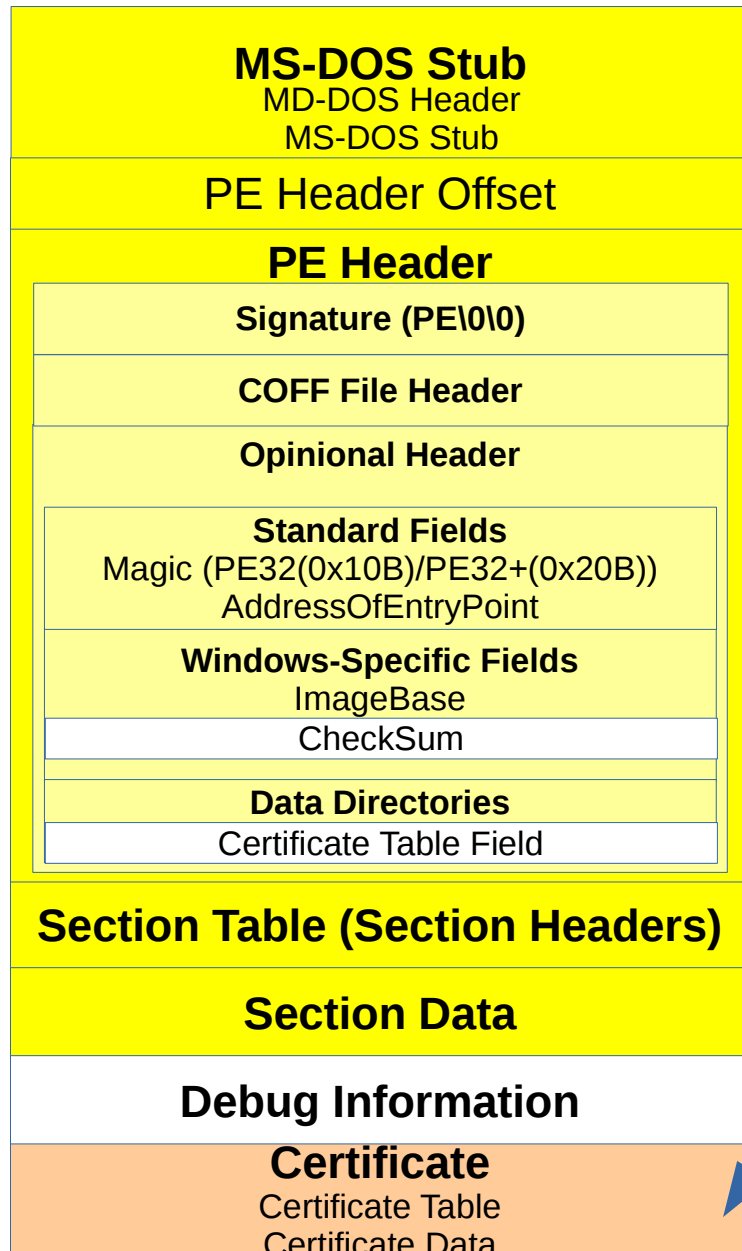
3. Signing server ASN.1 DER Encoding



Yellow regions: Data hashed to generate the unsigned digest.

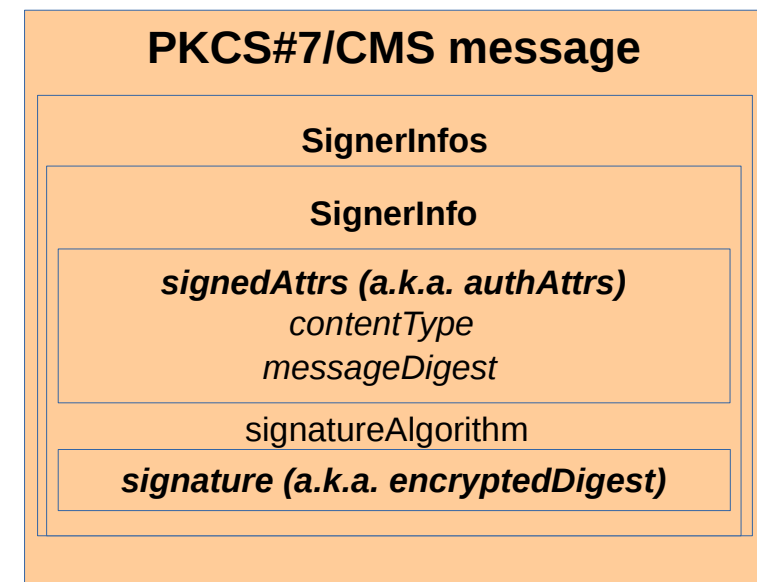
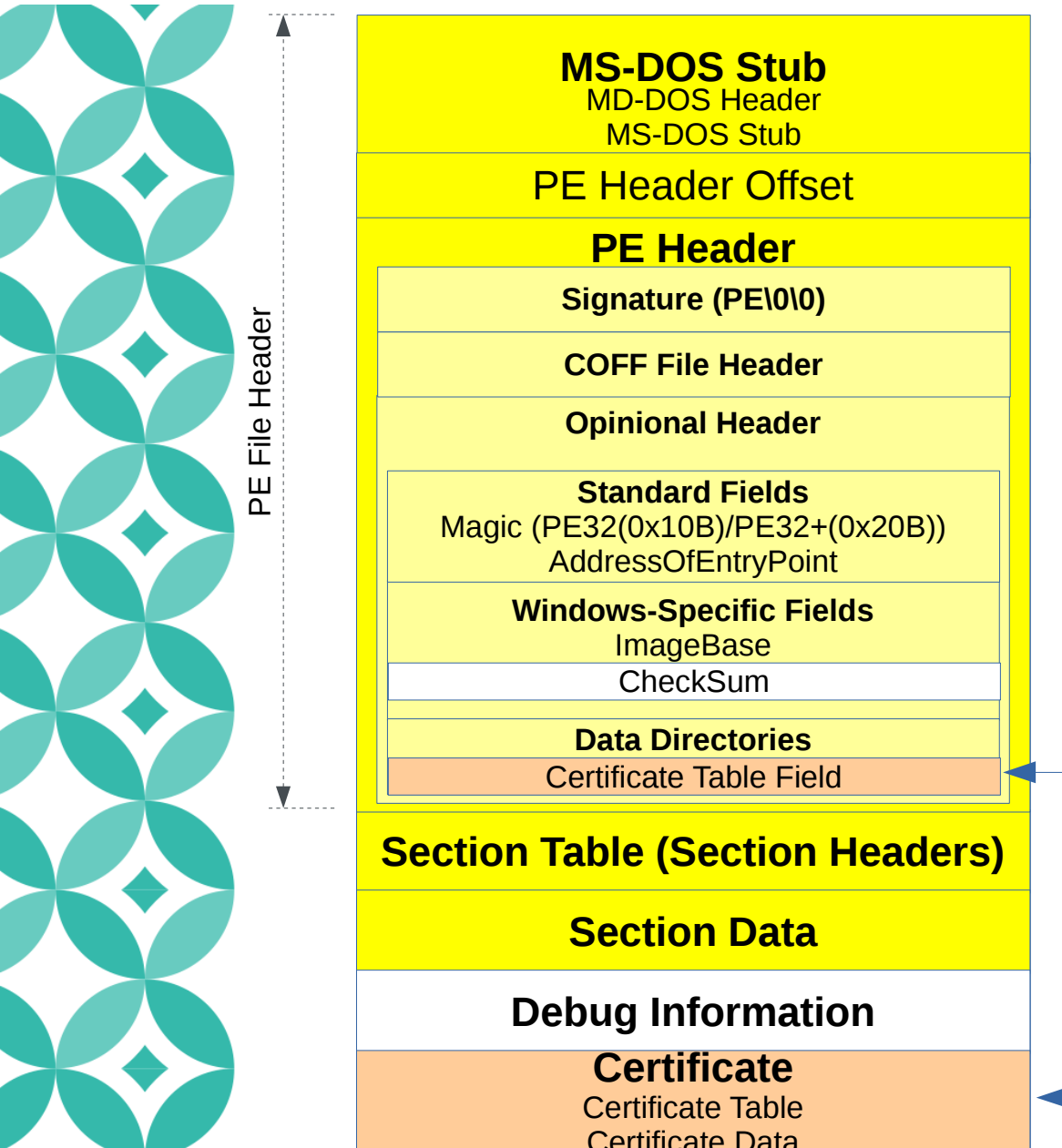


PE File Header



4. pesign in pesign-
repackage.spec

Yellow regions: Data hashed to
generate the unsigned digest.



5. pesign in pesign-repackage.spec

Yellow regions: Data hashed to generate the unsigned digest.

References

- Microsoft Portable Executable and Common Object File Format Specification
- Windows Authenticode Portable Executable Signature Format
- Unified Extensible Firmware Interface (UEFI) Specification Release 2.10 Errata A
- RFC5652 Cryptographic Message Syntax (CMS), September 2009

Thank You



openSUSE®
Asia Summit 2026